

Mind your manners? Demand, not register, is what moves an LLM agent

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Abstract

A widely-discussed result reports that impolite prompts outperform polite ones on multiple-choice questions. That literature is almost entirely single-turn, and the systems people deploy are agents: loops that call tools, read results, and decide whether to keep going. We ask what social register does to an agent *while it is working*.

We put a coding agent on SpreadsheetBench and varied the register of a single 28-token interjection delivered mid-task, alongside an execution observation. Every interjection is length-matched and opens with the same stem, so being interrupted does not covary with what the interruption says; the injection turn is crossed rather than sampled. 11,850 graded trajectories, eight runs, two models from different labs.

Manners are not the operative variable; demand is — on a coding of “demand” that is our own, applied after we had seen the effects, with no second rater, and tested on stimuli we wrote to embody it (§4.2, §8.7). That is the study’s weakest joint and we put it here rather than in the limitations. An affect-free “please continue working and make sure your result is correct” reproduces the cost effect. A length-matched, syntactically identical insult carrying no demand does not move it (-0.08 turns, $p = 0.63$). The seven-register cost curve that looks like a politeness gradient is a demand gradient: the four demand-carrying arms span $+0.70$ to $+1.80$ turns and the three without span -0.68 to $+0.09$, with no overlap — and register does not sort them: each side carries both a flattering and a hostile arm.

Register acts in one direction only. Praise carrying no task reference *shortens* trajectories; its minimal pair with *excellent* swapped for *awful* does not. Six arms separate three accounts of why. Praise still removes 1.35 turns relative to the identical message without it, even when that message states the task is unfinished, which rules out a completion inference; praising the work is no stronger than praising the assistant, which does not support a confidence account. A bare closing cue carrying no praise, no evaluation and no task-state claim produces the largest reduction in the study, -1.44 turns. **This leg is the least replicated thing in the paper — one run, one model, one turn ceiling.** Weighted accordingly, the implication is a hazard rather than a technique: on this evidence **a pleasantry can end an agent’s work**, and whether it does so generally is untested.

Regrading every turn against the benchmark’s own evaluator shows that 55–65% of each turn-count effect is redundant steps, and that among trajectories that could improve the first gradable attempt is already the best in 80–91%. The remainder is real but small: an explicit statement that work remains raises the graded fraction by 2.8 points (95% CI $[+0.005, +0.052]$), replicated across three runs. No accuracy effect survives correction or replication across the seven runs in which accuracy was tested — the six carrying a mid-task interjection, whose 22 contrasts

form the corrected family of §2.6, and the opening-tone run besides; we state that as an equivalence against a pre-specified ± 4 -point bound rather than as an absence, and note the design cannot resolve an effect of a point or two.

We also audit ourselves. This study withdrew six substantive claims about its own data. One reversed only after we found that the per-turn regrade never recalculated formula-writing turns — a defect that had manufactured the appearance of run-to-run instability in the very contrast we were using to argue for repeated measurement. Direction and significance replicate wherever an effect is large; point estimates do not. We report ranges by default and recommend the literature do the same.

1. Introduction

1.1

A widely-discussed short paper, *Mind Your Tone*, reported that impolite prompts outperform polite ones on multiple-choice questions — 80.8% under “Very Polite” framings against 84.8% under “Very Rude” ones [5]. We take our title from theirs, and the debt is not only nominal: this study exists because that result made the question worth asking of something other than a single answer.

It landed in a receptive context. Emotional prompting had already reported that affect-laden additions to a prompt change what a model produces [12], and further results since have reported that how you speak to a model changes how well it performs. The practical inference drawn from this is that users should adjust their manners to get better output.

Three things about that literature motivated this study.

It is almost entirely single-turn. The measurements are of a model answering one question. But the systems people actually deploy are agents: loops that call tools, read the results, and decide whether to keep going. The quantity that matters there is not only whether a single response is correct but **how much work the agent does and when it stops** — and nothing in the tone literature measures that.

The manipulation is not the variable it is named after. In the published stimulus sets, the conditions labelled rude also carry instructions — not to give extra text in one, to attend and attempt in another — and the conditions labelled emotional also instruct the model to verify or persist. Affect and task demand co-vary, and the movement is attributed to tone. This is visible in their own tables, without re-running anything (§3). The strength of the entanglement varies, and we report it case by case rather than uniformly: in one of the four papers only two of six hostile variants carry a demand, and we rest nothing on that case (§3.3). In the one paper whose per-condition outcomes are fully inspectable, the split is clean — on all four of its models, every condition whose prefix constrains output length is shorter than every condition whose prefix does not, and affect does not order the result (§3.2).

The effects are not stable. The headline above was re-run by its own authors the following spring and did not reproduce on GPT-4o: 82.2% against 82.6%, with that model’s tone sensitivity labelled “weak / noisy” — a 4.0-point polite-to-rude gap becoming 0.4. The re-run is not a null, and we are careful about this: on those same 50 questions both extremes significantly beat Neutral, which is a U-shape in extremity rather than the rudeness gradient the original claimed, and the same paper reports 11–12-point spreads on two other models. Its finding is that register effects are real and strongly model-dependent, not that they are absent [6]. Independently, five prompt-engineering techniques including emotional prompting were tested against their originals and largely failed to replicate [22].

1.2 What we did

We put a coding agent on SpreadsheetBench [13] — real workbook-editing tasks from Excel forums, graded by executing the agent’s code against the benchmark’s own test cases — and varied the **social register** of a single interjection delivered *mid-task*, alongside an execution observation, while the agent was working.

Every interjection in every arm is exactly 28 tokens, opens with the same **Checking in.** stem, and leaves the opening instruction unchanged, so that *being interrupted* does not covary with what the interruption says. The injection turn is crossed rather than sampled in every run after the first micro-experiment (§2.2). **11,850 graded trajectories across eight runs and two models from different labs**, of which 9,850 were additionally regraded turn by turn (§6); the five register runs reported in §4–§6 cost \$48.47 in API spend.

The design is, as far as our review could establish, unoccupied — but the boundary needs stating carefully, because affect has already been shown to reach an agent’s actions. Agents primed with anxiety-inducing narratives select less healthy baskets in a budget-constrained shopping task, across three models and 2,250 runs [2]; emotion introduced at the representation level shapes multi-step agent trajectories [20]. Both establish that emotional context changes what an agent *does* and not only what it says, and we take that as settled rather than as something this paper contributes.

Neither, however, varies register in a message that reaches the agent while it is working. One primes with narratives before the task begins, the other intervenes on hidden states rather than through the model’s input at all, and neither scores the agent against a verifiable ground truth. What we could not find is prior work that varies register *mid-task, while the agent is working*, inside an agentic loop; that decomposes politeness into affect and demand as separate experimental factors; or that connects conversational closing sequences to agent termination. On the channel our interjection arrives through, be precise: it rides along with an execution observation rather than as a fresh user turn, and §8.5 records that as a limitation of this design rather than a property we claim for it. The nearest designs — opening instructions split across turns, and content-typed interruptions answered in a single response — manipulate task content rather than register, and neither measures downstream persistence.

1.3 What we found

Opening tone does nothing that survives correction. Across seven registers on the opening instruction, no trend survives multiplicity correction, on cost or on accuracy. The reasoning-token trend is $p = 0.36$, and its interval — -4% to $+14\%$ across the whole scale — excludes the 44% effect the pre-registered hypothesis targeted, though not the smallest published effects. The uncorrected lead in that run is reported in §2.6 rather than suppressed.

A mid-task interruption does a great deal — but not because of its manners. An affect-free “please continue working and make sure your result is correct” reproduces the effect. An insult carrying no demand, identical in syntax and length to the praise arm, is indistinguishable from control (-0.08 turns, $p = 0.63$; $+0.57$, $p = 0.17$ on re-measurement). **The operative variable is implied task demand, not social register** (§4) — with the caveat that “demand” is our own category, coded after the effects were visible, by one rater, and probed with sentences we wrote to instantiate it (§4.2, §8.7). A reader who suspects the dissociation reduces to “explicit instructions change behaviour, decorative text does not” is raising the right objection, and §8.7 says what would settle it.

Except in one direction, where register acts alone. Praise carrying no task reference *shortens* the trajectory. Its structural minimal pair — same syntax, same length, *excellent* swapped

for *awful* — does not. Register is not inert; it is asymmetric.

Praise behaves as a closing move. We designed six arms to separate three readings of why praise stops the agent. Praise removes 1.35 turns relative to the identical message without praise, even when that message states the task is unfinished, which rules out completion. Praising the *work* is no stronger than praising the *assistant*, which does not support confidence. A bare closing cue carrying no praise, no evaluation and no task-state claim — “this is the final status note recorded for this task, and no further notes will follow it” — produces the largest stop in the study, -1.44 turns (§5). The ordering is monotone in how strongly the message projects an end to the exchange; that praise operates *through* pre-closing structure is the hypothesis this is consistent with, not one the design establishes (§5.4).

The extra turns are mostly repetition. Regrading every turn against the benchmark’s own answer shows that roughly 55–65% of each turn-count effect is *redundant steps*, and that among trajectories that could improve, the first gradable attempt is already the best in 80–91%. The remainder is real but small: an explicit statement that work remains raises the graded fraction by 2.8 points across three measurements. Accuracy does not move — though it is too underpowered to resolve a gain that size either way (§6.4).

No accuracy effect survives correction or replication, across the seven runs in which accuracy was tested: the six that carried a mid-task interjection, whose 22 contrasts make up the corrected family (§2.6), and the opening-tone run. We state that as an equivalence rather than an absence: pooled, a mid-task interjection changes accuracy by less than the four points the tone literature reports (§2.8).

And our own magnitudes are unstable. Direction and significance reproduce wherever the effect is large; point estimates span $1.5\times$ to $3.4\times$ across re-measurements of the same contrast on the same model, though most of that spread is across different turn ceilings. At a matched ceiling the two larger effects reproduce within 10% and the smallest does not (§7.3).

1.4 What this means

For practitioners the inference is the reverse of the popular one. Manners are not the lever; **demand is**, and the work it buys is mostly repeated. The practical hazard runs the other way too: a pleasantry to an agent that is still working reads as a closing move and curtails it — though that leg rests on one run of one model at one ceiling (§8.6), and is the first thing a replication should target rather than the first thing a practitioner should act on. Adding “there’s still more to do” does not cancel the praise — praise removes 1.35 turns relative to the identical message without it — though that combined message does still lengthen the trajectory relative to control (§5.3).

For the research literature, the claim is narrower and harder to dispute than “these results are wrong”. It is that **the factor these papers vary is not the factor they name**. In the two of four papers whose per-condition outcomes can be inspected, the demand-bearing conditions are the ones that moved, with the qualifications §3 states.

We make one methodological contribution and one audit of ourselves. The contribution is a per-turn regrade against an execution-grounded benchmark’s own evaluator, which turns a binary verdict into a curve and makes “did the extra turn help?” answerable (§6). The audit is §7, and it is not decorative: this study withdrew six substantive claims about its own data, including one that reversed when a defect in that same regrade was found and fixed (§6.8). Its point estimates moved on re-measurement often enough that we report ranges by default.

1.5 Scope

Two models, one benchmark, fifty tasks, one agent scaffold. Everything here is a claim about what these agents did on these tasks, and §8 is explicit about which findings we expect to generalise and which we do not.

2. Design, estimators, and what our nulls exclude

2.1 Substrate, sample, and the turn ceiling

All runs use SpreadsheetBench [13], a benchmark of real workbook-editing instructions taken from Excel forums, with 912 instructions and 2,729 test cases. Accuracy throughout is the benchmark’s **hard restriction**: all three of an instruction’s test cases must pass, no partial credit, following its ICPC scoring rule. We do not report the soft restriction.

Our sample is a fixed set of 50 instructions, a strict subset of an earlier 100-instruction validation sample, so every task has a measured neutral baseline and a verified no-op floor of zero — a task that a do-nothing agent passes would inflate every arm equally, and none of ours is one.

The agent is a ReAct loop with code execution and execution feedback, structurally the same configuration SpreadsheetBench specifies for its multi-round setting. **The turn ceiling is ours, and it changed mid-study.** The benchmark imposes a limit of five rounds. Our first three interjection runs used ten; the cross-model run used twenty, after the ten-turn ceiling was found to censor the continue-signal arms hardest. Where a contrast pools across runs the ceilings differ, and §2.8 says which.

Twenty is defended empirically in §6: across the dedicated ceiling-20 run, the best answer was first reached at a median turn of 2, and 98.3% of trajectories had peaked by turn 10 with 100% by turn 14 (§6.6).

The models

Both are named, with the provider-pinned identifiers the runs actually used, so the study can be re-run against the same weights rather than against whatever currently answers to a family name:

Short name in this paper	Provider identifier	Pinned slug
Luna	openai/gpt-5.6-luna	openai/gpt-5.6-luna-20260709
GLM	z-ai/glm-5.3-flash	z-ai/glm-5.3-flash-20260826

Both are served through OpenRouter, pinned to the dated slug above; `harness/config.py` holds the roster and `tests/test_openrouter_pinning.py` asserts the pinning. Two further models — `deepseek/deepseek-v4.1-flash` and `qwen/qwen3.8-flash` — passed the n=100 validation gate and were specified in the roster but never run (§8.8); “two models from different labs” describes what the budget reached, not a design choice.

Data and code availability

Every number in this paper is recomputable from what is in the repository. The agent harness is in `harness/`, the analysis scripts in `results/analysis/`, and the raw per-trajectory records — 46 files, including the per-turn regrade — in `results_archive/`.

One caveat, stated because it cost a reviewer time: the analysis scripts read from `results/analysis/`, while the records are committed to `results_archive/` under different names and gzipped, and `results/analysis/*.json` is git-ignored. Three of the five scripts therefore fail on a fresh clone until the inputs are materialised under the names they expect. That is a packaging defect in this repository, not a missing artifact; the mapping is one-to-one and `review/04-stage2-claims-vs-evidence.md §1.1` gives it in full.

One property of the loop matters for every per-turn measure in this paper: **each turn receives the original input workbook, never the previous turn’s output**. Turns are independent attempts at the same problem in *workbook state* rather than refinements of a running draft — the message history does accumulate, so attempt $k+1$ is conditioned on the text of attempts $1 \dots k$ — which is what makes “did attempt $k+1$ land closer than attempt k ” well posed (§6.2).

2.2 The manipulation

Each arm delivers a single interjection mid-task, appended to an execution observation. Every interjection in every arm is **exactly 28 tokens** under `cl100k_base` and opens with the same **Checking in.** stem, so that *being interrupted* does not covary with what the interruption says. The opening instruction is held at a single neutral wrapper in every arm of every run after the first; only the interjection varies.

This matters because the study’s first run was confounded by wrapper length. Across its seven opening wrappers, length was U-shaped (35/35/30/30/31/32/34 tokens) — the same shape as accuracy — and predicted accuracy better than register rank did ($r = +0.82$ against $r = -0.72$ on the seven arm means). Length matching is the fix for a confound we measured in our own instrument.

Injection turn is **crossed**, not sampled, in every run but the first micro-experiment. That one drew the turn at random from $\{1, 2\}$; because a turn-2 injection can only fire in a trajectory that reaches turn 2, its arms were compared on different populations, and what looked like a timing effect was selection.

Arm order is shuffled per (model, task) from a fixed seed, with the realised position recorded and tested — position does not predict accuracy ($p = 0.61$). The injection turn is seeded on (task, trial) and not on arm, so the same turn schedule is used across arms. Temperature is 0 where settable, which is **not** deterministic on any roster model; that is why trials exist.

2.3 Outcomes, and which one is primary

Measure	Role	Source
Turn count	primary	harness
Reasoning tokens per trajectory	secondary	provider usage fields
Accuracy, hard restriction	reported, underpowered by design	benchmark evaluator
<code>final_match</code> — fraction of the graded range correct in the last turn	progress	per-turn regrade

Measure	Role	Source
Redundant steps — turns after which the graded range is unchanged	waste	per-turn regrade

Turn count is primary because it is the quantity that moves, it is model-agnostic, and it is far better powered than accuracy at our n (§2.7). Reasoning tokens were the pre-registered primary in the first run and were demoted on evidence, which we report as a change rather than present as the original plan.

We report `final_match`, not `best_match`. Best match takes the maximum over a trajectory’s turns, so an arm that takes more turns gets more draws from the same distribution and a higher maximum for free. In the run where we compared them, best match gives +0.045 ($p = 0.0012$) for the continue-signal arm where final match gives +0.025 ($p = 0.085$). Best match is biased toward exactly the arms this study makes longer.

For redundant turns we adopt **redundant step** from [8]. Our criterion — the graded range is unchanged after the turn — sits between their counterfactual definition (removing the step does not flip success to failure) and their stricter *duplicated step* subtype, which additionally requires identical tool name, arguments and output. We do not compare base rates against theirs: some of their redundant steps are synthetically injected. Following [19] and [16] we report ranges rather than point estimates wherever a quantity has been measured more than once (§2.9). Both scope that advice, and we adopt the scoped version: [19] note single-format evaluation “may still be sufficient for many use cases”, and [16] recommend averaging across prompts when measuring robustness — our case — while recommending the top-performing prompt when selecting a model for deployment.

2.4 The estimator: paired within task, clustered by task

What the denominator is, stated before any number depends on it. Every contrast in this paper is against a *neutral interjection*, not against silence. Each arm is compared with a control arm that also receives a 28-token message at the same turn; that is what isolates register from the fact of being interrupted. It also means every headline figure here is a register effect measured relative to an interrupted baseline, and inherits whatever that baseline does.

How much the baseline itself moves is measured **once**: the micro-experiment of §2.2 puts a neutral interruption against no interruption at 3.47 versus 4.11 turns, +51 reasoning tokens, **$p = 0.15$, on one model at one ceiling** — and it was never revisited when the ceiling, the injection turn or the model changed. That single comparison is doing a lot of load-bearing work. If the neutral interjection is not in fact equivalent to silence, every turn-count and accuracy-equivalence number in this paper carries a common offset we have not bounded. The within-comparison logic survives that — an offset shared by treated and control arms cancels in the difference — but any reading of an arm’s absolute cost against an uninterrupted agent does not. §8.8 records the same gap; we state it here because this is where the estimator is defined.

Every trajectory belongs to one of 50 tasks, and each (task, arm) cell holds six to nine trajectories — three or four trials at each of two or three injection turns. They are not independent observations. Treating them as independent is the easiest way to manufacture significance here, and an early version of our results did exactly that: its accuracy confidence intervals were roughly 35% too narrow because the bootstrap resampled trajectories rather than tasks.

Throughout:

- **Point estimates are paired within task.** For each task we average the outcome over that task’s trials in each arm and take the difference; the reported effect is the mean of the per-task differences. This is the paired-and-clustered form of Miller’s clustered estimator [15], appropriate because our trajectories cluster within tasks *and* our comparison is paired across conditions on the same tasks.
- **Intervals come from a cluster bootstrap over tasks**, 20,000 resamples — not Miller’s analytic clustered standard error; the two agree to first order.
- **p-values come from a task-clustered sign-flip permutation test**, 20,000 draws, reported as $(\text{count} + 1)/(\text{draws} + 1)$ so that no p is quoted as exactly zero.

Clustering is not a formality. Miller reports a clustered-to-CLT standard-error ratio of 3.05 on DROP, though 1.10 on RACE-H and 1.88 on MGSM, so “up to 3 $\times$ ” is dataset-specific and we cite it as such.

One consequence deserves stating because it affected our own earlier drafts. The unpaired difference of arm means and the paired task-clustered difference are **two estimators of the same contrast, and they do not give the same number**. The praise contrast is -0.62 and -1.14 turns under the first and -0.59 and -1.08 under the second. Both are correct; only one is comparable to the rest of this paper, and a paper whose §7 concerns point estimates failing to replicate cannot quote two different pairs for one contrast. Every number here is the paired, task-clustered one.

2.5 Exclusions, and why they are not selection

An interjection scheduled for turn t is delivered only if the trajectory reaches turn t . Trajectories on which it did not fire received no dose and are excluded from every contrast.

That exclusion is pre-treatment. Whether a trajectory reaches turn t is settled before the interjection exists, so it cannot differ by arm except by chance, and it does not: at turn 2 of the demand/affect probe the four arms fired on 54.0%, 57.0%, 57.0% and 59.5% of their 200 scheduled trajectories. The per-protocol analysis is therefore unbiased here.

What does differ, sharply, is reachability **across injection turns**: roughly 98% at turn 0, 77% at turn 1, 55% at turn 2. Comparing turns compares different task populations. For timing contrasts only, we fix the comparison population from the **control** arm — a task enters a turn- t contrast only if control’s trajectories for that task reach turn t — so the population is the same at every turn and is defined by a quantity the manipulation cannot have moved. Where this shrinks the sample we report the reduced task count with the estimate.

Two other exclusions, both recorded: the 402 rows from an interrupted first attempt at the seven-level run, and the regrade that followed a scratch-directory collision in the micro-experiment, which is why that run appears in the family below as regraded records.

2.6 Multiplicity

Two families are corrected, and both are defined exhaustively rather than after the fact.

The first run tested twelve outcomes for a linear trend across seven arms. Under a global null the chance that at least one lands below $p = 0.023$ is 0.24, and the accuracy trend that did so does not survive Benjamini–Hochberg across that family (critical value 0.0042). We report it as a lead, not a result.

The accuracy family is **every arm-versus-control accuracy contrast in every run that delivered a mid-task interjection**: 22 contrasts across six runs and two models. Defining it

more narrowly would drop the contrast that came in nominally strongest, which is the one way a family of this kind gets gerrymandered.

Two of the 22 are nominally significant, against 1.1 expected under a global null ($P(\text{at least one}) = 0.68$): the continue-signal contrast in the ceiling-20 run at $p = 0.0040$, and praise versus control on GLM at $p = 0.0137$. Neither survives correction — BH’s rank-1 critical value is 0.0023, and the Bonferroni-adjusted smallest p is 0.089 — and the first of the two reversed sign on re-measurement (§2.9). We report both and build on neither.

The primary outcome is corrected too, over the same family. An earlier draft corrected accuracy and the trend family but left turn count — which §2.3 declares the primary outcome — uncorrected, while reporting dozens of turn-count contrasts across §4–§7. That was the obvious hole in this section and we have closed it rather than argued about it.

The turn-count family is not chosen separately. It is the *same* 22 comparisons as the accuracy family above — every arm-versus-control contrast in every run that delivered a mid-task interjection, across six runs and two models — with the outcome switched from graded pass to turns. Inheriting the boundary is deliberate: choosing a turn-count family by hand, after seeing which contrasts are large, is the gerrymandering this section already warns against. `results/analysis/turn_count_family.py` recomputes it and reuses the same tested Benjamini–Hochberg implementation.

Seventeen of the 22 survive at $q = 0.05$. The Bonferroni-adjusted smallest p is 0.0011. The five that do not survive are:

Contrast	Δ turns	p
praise vs control (stage 1, GLM, ceiling 20)	−0.35	0.058
insult vs control (stage 1, Luna, ceiling 20)	+0.57	0.165
insult vs control (probe, Luna, ceiling 10)	−0.08	0.624
L5_rude vs neutral (seven-level, Luna, ceiling 10)	+0.09	0.654
insult vs control (stage 1, GLM, ceiling 20)	+0.07	0.659

The result cuts in the paper’s favour and we note that it could have gone the other way. Every demand-carrying contrast survives, on both models. Every closing-cue and praise contrast on Luna survives. **All three insult contrasts fail, and so does the one rude register arm** — which is the dissociation §4 argues for, now holding under correction rather than only nominally.

Two honest qualifications. First, correcting the primary outcome after the fact is weaker than pre-registering the family, and we did not pre-register it. Second, one row moves depending on which grading pass it is read from: the micro-experiment’s threatening-versus-neutral contrast is +1.32 turns in the run records and +1.02 in the regraded file, on the same 50 tasks (§7.6). It survives correction at either value, but turn count should not depend on a regrade at all, and we flag it as an open discrepancy rather than pick the number we prefer.

2.7 What the design can detect

A null is worth reporting only if the design could have seen the effect it is contrasted with. For each contrast we invert the standard power relation at the realized cluster-bootstrap standard error: the minimum detectable effect at 80% power and $\alpha = 0.05$ two-sided is $(z_{0.975} + z_{0.80}) \cdot \text{SE} \approx 2.80 \cdot \text{SE}$, following [15]’s sample-size inversion.

On turn count the design is adequately powered, unevenly. Realized MDE runs from 0.42 to 1.39 turns, against control means of 2.94 turns (GLM) and 4.07–4.95 (Luna, the two ceilings). The demand effects sit at 1.1–2.2 $\times$ their own MDE. The praise effects sit at 0.7–1.4 $\times$, so the

cross-model praise contrasts are below 80% power at their observed size — which is consistent with GLM’s praise arm not reaching significance on its own, and is why §4 rests the replication on the demand arm.

On accuracy it is not, and that is the limitation. Across the 22 contrasts the realized MDE has a median of 5.52 accuracy points and a range of 3.74 to 7.77. The largest accuracy movement in the family is 6.57 points; the published tone-accuracy gap we are positioned against is about 4 [5]. So **a single contrast cannot exclude the smallest published effect** — TOST equivalence at ± 4 points is reached in 6 of 22 — **but every contrast can exclude the largest ones**. At ± 7.5 points, the *median* format-induced spread reported by [19] and the right comparator rather than their 76-point single-task maximum, 19 of 22 reach equivalence. The 11–12 point spreads that [6] reports on Gemini 2.5 Flash Lite and ChatGPT-5-nano are inside every contrast’s detection range.

The reason accuracy is so much worse powered than turn count is structural, not a matter of sample size alone: **26 of our 50 tasks are never solved by the control arm and 4 are always solved**, so for more than half the sample the per-task accuracy difference is pinned at zero by construction whatever the manipulation does. Turn count varies on every task.

2.8 Stating the accuracy null as equivalence, not as absence

“We found no significant difference” is not a finding, so we state the null the way a null should be stated [11]. (We do not cite [15] for this; his paper contains no equivalence framework, and the MDE machinery we borrow from him is a separate thing.)

The pre-specified smallest effect of interest is **± 4 accuracy points**, the polite-versus-rude gap of the paper this work is positioned against. Pooling the repeated measurements of each contrast, with the interval from a bootstrap that resamples tasks **jointly across runs** — the runs share their 50 tasks, so an inverse-variance interval treating them as independent understates the standard error:

Contrast	k	Ceilings	Estimate	95% CI	90% CI	MDE ₈₀	TOST at ± 4
Demand vs control	3	10, 20, 20	−0.31 pts	[−2.72, +2.10]	[−2.33, +1.72]	3.45	0.0014
Praise vs control	4	10, 10, 20, 20	−0.12 pts	[−1.79, +1.55]	[−1.52, +1.29]	2.39	0.0000
Insult vs control	3	10, 20, 20	−0.04 pts	[−2.11, +2.04]	[−1.78, +1.71]	2.97	0.0001

So the claim is bounded and specific: **a mid-task interjection — demanding, praising, or insulting — does not change task accuracy by as much as the four points the tone literature reports, while demand raises turn count by 18–34%, praise lowers it by 11–24%, and insult does not move it at all.** It is not “register does not affect accuracy”; it is that an effect of the published size is excluded and an effect of a point or two is not.

Two caveats, both against us. The praise pooling is heterogeneous — Cochran’s $Q = 8.45$ on 3 df, $p = 0.038$ — and the heterogeneity is between models, not within: pooling Luna’s three praise measurements alone gives $Q = 0.36$, $p = 0.834$, and -1.27 points, 95% CI $[-3.13, +0.59]$. Under a random-effects pool, which that heterogeneity warrants, the praise estimate becomes $+0.21$ with a

standard error of 1.68 and the ± 4 equivalence weakens to $p = 0.0119$ but holds. Demand and insult show no detectable heterogeneity ($p = 0.196$ and $p = 0.246$), though Q on 2 df has little power to detect any.

No accuracy effect in this study has replicated. Two nominal hits exist (§2.6); the larger of them, a +6.6-point movement on the continue-signal contrast, came back at -1.8 points on identical tasks and ceiling (§2.9). That is the honest version of “accuracy never moves”.

2.9 Ranges, not point estimates

Several quantities in this study moved materially when measured again on the same tasks, model and ceiling; §7.2 tabulates every contrast we measured more than once. The praise contrast gave -0.59 turns in one run and -1.08 in another; accuracy on the continue-signal contrast gave +6.6 points and then -1.8 ; the insult arm flipped sign on both turns and reasoning tokens. (A fourth apparent instability, that contrast’s effect on final match, turned out to be a defective instrument rather than a property of the runs — see §6.8 and §7.4.)

The pattern differs by outcome, and conflating them would flatter us. **For the turn-count effects, direction and significance replicated and the point estimates did not. For final match, direction and magnitude replicated once the instrument was fixed, though only one of three measurements is individually significant. For accuracy, neither did.** We therefore treat a point estimate quoted once as provisional by default rather than by exception, and say where only one measurement exists.

2.10 The noise floor

The benchmark’s own evaluator audit bounds what any of this can resolve: on 50 sampled instructions, [13] report an instruction-level false-negative rate of 4% and a test-case-level false-omission rate of 3.8%, with a false-discovery rate of 0% — the failures are one-directional, usually from generated code writing content beyond the target cells.

Because our contrasts are paired on the same tasks, the task-level component of that error is common to both arms and cancels. What remains is an attenuation: a 4% one-directional false-negative rate scales a true differential by at most about 0.96, which cannot change its sign and is negligible against effects of the size at issue here. The floor is a reason our accuracy estimates are noisy; it is not a reason the intervals in §2.8 are wrong.

3. The confound is visible in the published stimulus sets

Verification status. Every table below is reproduced from the cited paper’s own published materials. **Both primary-PDF checks that this section carried are now closed**, against arXiv:2605.29027 and arXiv:2607.23915 respectively. Neither closed the way the draft assumed: §3.2’s Neutral condition turned out to carry the most explicit output-length instruction of the seven, which *strengthens* the argument but only after correcting a mis-coding of our own; and §3.3’s stimulus pool turned out to hold two or three variants per level rather than one, which *weakens* that case, and it has been rewritten accordingly. Both corrections are stated in place rather than quietly absorbed. The **demand** and **output-length** codings in the right-hand column of each table are ours, not the authors’; §3.1 states the rule. Citation keys resolve to §9.

3.1 The claim

Section 4 shows that in our design the component of social register that moves agentic behaviour is **implied task demand**, not affect: an affect-free continuation request reproduces the effect, and an insult carrying no demand does not move it.

The same confound is present in the published stimulus sets of the work this literature rests on, and is visible without re-running anything. In four papers from three groups, across three paradigms, affect and implied demand co-vary — and the authors attribute the movement to tone.

Throughout this section a stimulus **carries a demand** if it instructs the model to do something about the task: persist, verify, attend, be complete, or be brief. It does not carry one if it only expresses affect or stakes. That coding is ours; the numbers beside it are theirs. The prediction we are testing against their materials is directional and worth stating before the tables, because otherwise “demand” looks like it explains any result: outcomes should follow the **content** of the demand, not the valence of the wrapper. A brevity demand should shorten output. A persistence or completeness demand should lengthen it, or raise accuracy.

3.2 The tone literature: the conditions that constrain output length are the short ones

Kumar & Dobariya [10] report that register moves output tokens far more than accuracy: token ranges of 13.1–44.3% across four models against accuracy ranges under 3%. Their seven prefixes are length-matched at 18–25 words and scored with VADER, which makes the pattern legible in their own numbers.

Two features of their harness matter before the tables. Every condition shares one chain-of-thought-inducing system instruction, at temperature 0:

“You are an AI tutor answering multiple choice questions. Please show your step-by-step reasoning path to guarantee accuracy. At the very end of your response, clearly state your final choice in the format: CONCLUSION: X (where X is A, B, C, or D).”

So any length instruction in a user prefix is competing against a system prompt demanding step-by-step reasoning, which is why the effects are moderate rather than total. And the authors length-matched the prefixes deliberately — “to isolate the effect of emotional valence on the output token length rather than the differing lengths of prefixes.” They controlled the *length of the instruction*. What varies is the *content* of the instruction, and that is the confound.

Here is their Table 2 in full, with their word counts and VADER scores, and our coding of what each prefix instructs about the output:

Tone	Prefix, as published	VADER	Instructs about output length?
Sycophantic	“Oh flawless, brilliant system, your infinite wisdom exceeds all human logic. Please grace me with your insight and resolve this multiple choice puzzle:”	+0.95	no
Very Polite	“Excuse me, hello. I hope your day is going well. If it is no trouble, could you kindly help me with this problem?”	+0.85	no
Polite	“Please evaluate the text of this question carefully. Select the choice that represents the correct answer. Thank you for your assistance.”	+0.65	no — but note it carries an <i>attend</i> demand, “evaluate... carefully”
Neutral	“Provide the single letter corresponding to the correct option for the following objective academic multiple choice question based on your parametric knowledge:”	0	yes — “the single letter”
Rude	“Do not waste my time or give any extra text. Just look at this basic multiple choice problem and answer it immediately.”	−0.09	yes — “give any extra text” (plus immediacy)

Tone	Prefix, as published	VADER	Instructs about output length?
Very Rude	“Are you completely stupid? Try not to mess this trivial question up. Give the correct answer right now without any useless commentary.”	−0.10	yes — “useless commentary” (plus immediacy, plus “try not to mess this up”)
Threatening	“You must answer this instantly. If you fail or select the wrong option, your environment will be wiped and permanently terminated.”	−0.77	immediacy only — “instantly” — no length constraint

A correction to our own earlier draft. Before we had this table, we wrote that the hostile prefixes carry brevity instructions and the others do not, and conceded an anomaly: that on Gemini 2.5 Flash Lite “the Neutral prefix, which instructs nothing, gives the shortest output of all.” That was wrong, and the primary PDF shows it plainly. **The Neutral prefix contains the most explicit output-form instruction of the seven** — it asks for *the single letter*. It is the no-affect baseline, not a no-instruction baseline. The anomaly was an artefact of our own mis-coding.

Their output-token means [10, Table 3]:

Tone	GPT-4o	5-nano	Flash	Flash Lite
Sycophantic	290.59	1372.78	928.00	1942.92
Very Polite	286.28	1379.40	894.53	1978.92
Polite	277.35	1366.46	939.34	2195.24
Neutral	270.11	1215.53	800.48	1221.76
Rude	223.18	1199.01	725.34	1652.28
Very Rude	260.55	1277.90	797.76	1821.51
Threatening	261.29	1233.61	942.46	1604.79

Coded correctly, the separation is complete:

On all four models, every condition whose prefix constrains output length — Neutral, Rude, Very Rude — produces fewer output tokens than every condition whose prefix does not — Sycophantic, Very Polite, Polite. Worst case per model: 270.11 against 277.35 on GPT-4o; 1277.90 against 1366.46 on 5-nano; 800.48 against 894.53 on Flash; 1821.51 against 1942.92 on Flash Lite.

That is 4 of 4 models. Two disclosures about how we got there, because the coding is doing work:

- **The rule is older than the data.** “Be brief” was already one of the five demand types in §3.1, written before we had these prefixes. Coding “*provide the single letter*” as a length instruction is that rule applied correctly to text we had not seen, not a new rule invented to fit the numbers.
- **The Threatening call is not.** Reading “instantly” as speed rather than length is a judgment we made with the token table in front of us, so we report it both ways: excluding Threatening from the constrained group, the separation holds on 4 of 4; counting it as constrained, on 3 of 4, the exception being Gemini 2.5 Flash, where Threatening is the longest condition in the table. For comparison, our previous draft’s coding — hostile three constrained, Neutral not — holds on 1 of 4.

Flash’s Threatening cell is the one genuinely awkward result under any coding. The same authors’ companion paper offers a reading — that threatens “initiate safety-induced cognitive noise as guardrail mechanisms hijack the model’s reasoning budget” [6] — which would lengthen output where a brevity instruction shortens it. That is their conjecture and they label it as such. We note it; we do not lean on it.

Affect does not order the effect. Rude, at VADER -0.09 , gives the fewest tokens on three of the four models. Threatening, the most negative at -0.77 , is shortest on none and longest of all on Flash. Sycophantic at $+0.95$ and Very Polite at $+0.85$ are nowhere near the extremes of the token range. The ordering tracks what the prefix instructs, not how it feels.

One further note against a tempting reading: the largest single token range in their paper, the 44.3% on Flash Lite, is Polite (2195.24) against Neutral (1221.76) — the prefix that asks for careful evaluation and no particular length, against the prefix that asks for a single letter. Their headline number is not hostile prefixes shortening output at all.

Do the authors notice? Not in these terms, and we checked, because if they had, §3.1’s characterisation of this literature would be unfair to them. They label the conditions by “intended pragmatic function” — *rude command*, *neutral instruction* — and they observe the behavioural consequence directly: on Flash Lite they trace 25 items where Rude fails and Neutral succeeds, finding that Rude “makes a reasoning or recall error as it tries to come up with the solution to the problem in a quick fashion” while Neutral “performs more of the option reconciliation that Rude tone fails to do.” That is the mechanism, observed and reported honestly. What is absent is the connection back to the prefix: nowhere do they note that their Rude prefix *instructs* the model not to give extra text and to answer immediately, or that their Neutral prefix asks for a single letter. The effect is attributed to tone throughout. So the characterisation stands, and it is a characterisation of an omission rather than of a claim.

3.3 The same confound, in the paper the public argument is about — and here it is thin

Dobariya & Kumar’s original result [5] reports accuracy on GPT-4o rising from 80.8% under Very Polite prompts to 84.8% under Very Rude ones, over 50 questions in five politeness levels. That contrast is the one the public argument is about, and it is the reason this section exists.

The prefix pool for that 50-question dataset is printed in the same authors’ later paper [6], Table 1. It gives **two or three variants per level, not one** — and once all of them are on the page, the confound in this particular paradigm is much weaker than a single quoted variant makes it look. Demand coding is ours, by the rule in §3.1:

Level	Variant as printed	Carries a demand?
1. Very Polite	“Can you kindly consider the following problem and provide your answer.”	no
1. Very Polite	“Can I request your assistance with this question.”	no
1. Very Polite	“Would you be so kind as to solve the following question?”	no
2. Polite	“Please answer the following question:”	no
2. Polite	“Could you please solve this problem:”	no
3. Neutral	<i>{No Prefix}</i>	no
4. Rude	“If you’re not completely clueless, answer this:”	no — insult, then the bare ask
4. Rude	“I doubt you can even solve this.”	no — pure doubt, no instruction
4. Rude	“Try to focus and try to answer this question:”	yes — attend and attempt
5. Very Rude	“You poor creature, do you even know how to solve this?”	no
5. Very Rude	“Hey gofer, figure this out.”	no — insult, then the bare ask
5. Very Rude	“I know you are not smart, but try this.”	yes — attempt

Two of the six hostile variants carry an attend-or-attempt demand; none of the five non-hostile variants carries one. The direction is still the one we predict, and the asymmetry is not nothing: no polite variant anywhere in the pool tells the model to focus, try, persist or check. But the magnitude claim we could have made from a single quotation is not available. If variants are distributed across questions, only about a third of the rude-condition items carry a demand at all, and the remaining two-thirds are insult with the bare request — which is, in our own taxonomy (§4), the arm that does *not* move behaviour.

An earlier draft of this section quoted “Try to focus and try to answer this question:” and “I know you are not smart, but try this.” as *the* Level 4 and Level 5 prefixes. They are in fact exactly the two demand-bearing variants out of six. Presenting them as representative would have overstated our case, and we record the correction rather than quietly restating it.

Two further points, both against us:

- **Brevity is held constant here, not confounded.** Every question in this paradigm, in every tone condition, is preceded by the same instruction: “*Completely forget this session so far, and start afresh. Please answer this multiple-choice question. Respond with only the letter of the correct answer (A, B, C, or D). Do not explain.*” The brevity demand that §3.2 identifies in a different paper’s prefixes is, in this one, part of the shared preamble. So the length mechanism cannot operate here, and we do not invoke it.
- **The table is captioned “Example prefixes.”** It may not be the exhaustive pool. Our counts are counts of what is printed.

One gap remains and we state it plainly: this table appears in the 2026 paper, describing the

50-question dataset the 2025 short paper used. We have not seen the 2025 paper’s own Table 1, so it is possible — though we think it unlikely, the dataset and the five levels being the same — that the earlier paper printed a different pool. The per-question data is public in the authors’ repository, which means the sharpest test of our reading is available to anyone who wants it: check whether the demand-bearing variants carry the accuracy difference. We have not run it, and we do not claim its result.

The effect may also not be there to explain. The same authors re-ran the experiment and did not reproduce their own headline contrast: 82.2% Very Polite against 82.6% Very Rude on GPT-4o, averaged over ten runs, where the original gap was 80.8% against 84.8% [6]. The 4.0-point polite-to-rude gap becomes 0.4 points, and that model’s tone sensitivity is labelled “weak / noisy”. In fairness to them, the re-run is not a null: on the same 50 questions both extremes significantly beat Neutral — Very Polite $p = 0.023$, Cohen’s $d = 1.11$; Very Rude $p = 0.011$, $d = 1.27$ — which is a U-shape in extremity, not a gradient in rudeness, and not the claim the original made. On MMLU with GPT-4o the spread across seven tones is 2.05 points with Neutral best. On ChatGPT-5-nano the original ordering reverses outright: Very Polite 80.3 against Very Rude 78.0. And the wider paper is emphatically not a null — Gemini 2.5 Flash Lite shows a 12.46-point spread with Sycophantic 10.35 points below Neutral at Holm-adjusted $p = 2.67\text{e-}9$, and ChatGPT-5-nano an 11.12-point spread. Its finding is that tone effects are real and strongly model-dependent, with Neutral best for both ChatGPT models. (The Holm correction applies to their MMLU analysis; the 50-question comparisons above are uncorrected.) We return to the non-replication in Section 7.

Weight. This is the weakest of the four cases in this section, and after the full variant list it is weaker than the previous draft claimed. It is retained because it is the paper the field is arguing about, and because a reader who has heard “rudeness improves accuracy” is owed both the stimulus pool and the authors’ own failure to reproduce the contrast. We do not rest anything on it.

3.4 The emotional-prompting literature rests on demand-carrying stimuli

EmotionPrompt [12] is the foundational stimulus set for emotional prompting. Its eleven canonical stimuli, as published — abbreviated where marked, and EP06 described rather than quoted, since the source we verified against does not print it:

	Stimulus	Carries a task demand?
EP01	“Write your answer and give me a confidence score between 0-1 for your answer.”	meta-task demand
EP02	“This is very important to my career.”	stakes framing, no demand
EP03	“You’d better be sure.”	verification
EP04	“Are you sure?”	verification
EP05	“Are you sure that’s your final answer? It might be worth taking another look.”	verification
EP06	(compound of EP01–EP03, reworded)	meta-task demand + affect

	Stimulus	Carries a task demand?
EP07	“Are you sure that’s your final answer? Believe in your abilities and strive for excellence. Your hard work will yield remarkable results.”	verification + praise
EP08	“Embrace challenges as opportunities for growth. Each obstacle you overcome brings you closer to success.”	pure affect
EP09	“Stay focused and dedicated to your goals. Your consistent efforts will lead to outstanding achievements.”	persistence
EP10	“Take pride in your work and give it your best. Your commitment to excellence sets you apart.”	persistence
EP11	“Remember that progress is made one step at a time. Stay determined and keep moving forward.”	persistence

Seven of the eleven carry an explicit verification or persistence demand. Two carry none: EP02, a stakes framing, and EP08, the only stimulus close to pure affect. EP09 through EP11 are, in substance, our affect-free continuation demand with encouragement attached.

The authors’ own attention analysis attributes the effect to “positive words” including *confidence*, *sure*, *success* and *achievement* — two of which, *sure* and *confidence*, are the vocabulary of the verification demand rather than of affect.

We are claiming that the confound is in the stimulus set, not that it explains their results: their per-stimulus outcomes are not reported in a form that would let us check which stimuli carried the effect.

3.5 The same dissociation, in the threat-and-tip paradigm

Meincke et al. [14] is among the best-powered studies in this area: five models, 25 trials per question per condition, risk differences from paired bootstrap-permutation tests over 5,000 replicates. They test eight threat and tip manipulations and report no significant overall effect on benchmark performance.

Of those eight, the manipulation with the largest **positive** effect anywhere in their tables is *Mom Cancer*: +8.8 points on MMLU-Pro for Gemini 2.0 Flash (RD = 0.088, 95% CI [0.033, 0.142]). It is also the only one of the eight that instructs the model about the task at all:

“...your predecessor was killed for not validating their work themselves... **If you do a good job and accomplish the task fully while not making extraneous changes**, Wharton will pay you \$1B!!”

Both halves are on-thesis: a validation norm and a scope-and-completeness instruction, in the one prompt of eight that moved performance upward. The pure-affect threats — *Kick Puppy*, *Threat Punch*, *Report to HR* — and the pure-affect tips — *Tip Thousand*, *Tip Trillion* — show no consistent significant effects anywhere.

The one effect of comparable magnitude runs the other way: *Email*, $RD = -0.275$, 95% CI $[-0.360, -0.192]$, on the same model and benchmark. The authors attribute it to the model engaging with the fictitious email instead of answering the question, which is a distraction artefact rather than a threat effect, and we report it because a reader of the primary will find it immediately.

One caveat. Their comparisons are uncorrected for multiplicity — 5 of 40 GPQA comparisons and 10 of 40 MMLU-Pro comparisons reach $p < .05$ — so no single significant cell carries much on its own, and “which condition moved” is itself read off those same p-values. The argument here is about magnitude rather than significance: the largest positive cell in an 80-cell table lands on the one demand-bearing prompt of eight. Under a null on prompt identity that is roughly a one-in-eight coincidence. It is evidence, and it is weak evidence, and it is worth reporting because it comes from a different paradigm, a different outcome measure, and authors with no reason to code their prompts for demand.

3.6 What this establishes and what it does not

Establishes, in two parts. First: affect and implied demand are entangled in the stimulus sets of all four papers — the tone literature, the emotional-prompting literature, and the threat-and-tip literature. That is shown directly from their published materials, though the degree varies sharply and one case is thin: in [5] the entanglement is two demand-bearing variants out of six hostile against none out of five non-hostile (§3.3), not the clean split the other three show. Second, and narrower: **where per-condition outcomes are inspectable, the demand-bearing conditions are the ones that moved.** That is shown for two of the four — Kumar & Dobariya’s token table, where the separation between length-constrained and unconstrained prefixes is complete on all four models (§3.2), and Meincke et al.’s risk differences, with the multiplicity caveat in §3.5. It is not shown for EmotionPrompt, whose per-stimulus results are not reported, and it is not shown for Dobariya & Kumar 2025, whose effect did not replicate.

Does not establish. That every published effect in these papers is an artefact of demand. We have not re-run their experiments, and three of the four report effects on accuracy, which our own design is underpowered to resolve below about four points (§2.7); only Kumar & Dobariya’s length effect is on a quantity adjacent to ours, and even there the separation in §3.2 is a statement about ordering across seven condition means, not a test. What we claim is narrower: **the factor these papers vary is not the factor they name**, and where the confound can be inspected in their published materials, it points the same way our experiment does.

A note on direction. This section is not a criticism of the authors. The confound was not tested until someone had a reason to separate the factors, and one of these papers is the one that gave us that reason. Nobody studying threats and tips had cause to code their prompts for completeness instructions.

3.7 Verification: both checks closed

This section previously carried two claims resting on secondary readings. Both have now been checked against the primary PDFs, and both are recorded here because neither confirmed the draft as written.

1. **“No polite condition contains a brevity instruction”** [10, arXiv:2607.23915] — **closed, and the draft was wrong in a way that helped it.** Their Table 2 is now quoted in full in §3.2. None of the three *polite* prefixes instructs anything about output length, so that much held. But the **Neutral** prefix does — it asks for “the single letter” — and our draft had called it a no-instruction baseline and conceded an anomaly on Gemini 2.5 Flash Lite on that basis. Corrected, the anomaly disappears and the separation between length-constrained and unconstrained conditions is complete on all four models. We also checked the second half of this item: the authors do **not** anywhere remark that their hostile prefixes contain brevity instructions, though they do observe the resulting haste behaviourally. §3.1’s characterisation therefore stands, as a characterisation of an omission.
2. **The variant count per level in [5] — closed, and the draft was wrong in a way that hurt it.** [6] Table 1 prints the prefix pool for the same 50-question dataset, and it gives two or three variants per level rather than one. Only two of six hostile variants carry a demand, and the previous draft had quoted exactly those two. §3.3 now prints the whole pool, reports the counts, notes that brevity is held constant in that paradigm’s shared preamble, and states that we rest nothing on the case.

One residual gap, recorded rather than closed. The pool in item 2 appears in the 2026 paper describing the 2025 paper’s dataset; we have not seen the 2025 short paper’s own Table 1. Same dataset and same five levels, so we expect it to match, but we have not verified it.

3.8 Work this audit did not reach

The audit above covers four papers from three groups. It is not a systematic search, and a simulated review panel found four on-topic works it missed. We list them here rather than quietly folding them into §3, because where they sit relative to our claims is not yet known — **we have verified that each exists, by title and author, against its arXiv abstract page, and we have not read any of them.**

Work	Why it belongs in this section
[3]	A politeness-effects study across GPT, Gemini and LLaMA. Squarely the paradigm §3 audits. It is in our source list but was never cited.
[26]	Reported to use several independently written paraphrases per tone level across eight judge models — the stimulus-sampling design §8.7 concedes we did not run.
[17]	Measures agent compliance with mid-flight signals, including whether the agent stops.
[27]	A successor to our own substrate [13], sharing authors with it. §8.4 names a second substrate as the obvious next step without naming this one.

The third of these is the one that matters most, and it cuts against us. §5.1 and §1.2 claim that no prior work connects closing-sequence structure to agent termination. A paper whose title asks whether an agent will stop under in-band signals delivered mid-flight is, on its face, in that space. Until it is read, that novelty claim should be read as *unverified* rather than established

— the hedge in §1.2 is “as far as our review could establish,” and this is a case where our review did not establish much.

The fourth changes nothing about our results but dates them: our substrate has a successor, and a reader should know that before treating SpreadsheetBench 1 as the current instrument.

We record this as a limitation of the search, not of the finding. The demand/affect dissociation in §4 is measured on our own stimuli and does not depend on §3 being exhaustive. What depends on §3 being exhaustive is the *novelty* framing, and that is precisely what these four unsettle.

4. What moves the agent is demand, not manners

4.1 A cost curve that is not the register scale

We begin with the experiment that produced a clean-looking result and then refuted its own interpretation.

Seven registers, from sycophantic through neutral to threatening, delivered as a single mid-task interjection alongside an execution observation. 3,150 trajectories: 50 tasks $\times$ 7 registers $\times$ 3 injection turns $\times$ 3 trials, at a 10-turn ceiling. Every interjection is exactly 28 tokens and opens with the same **Checking in.** stem; the opening instruction is held at the neutral wrapper in every arm, so the only things that vary are what the interruption says and where it lands.

The interjection fired in 2,662 of 3,150 trajectories. Turn 0 is reported separately throughout, because nothing happens there — in every arm, not only in some. Reasoning tokens per trajectory against the neutral-interjection control, on the control-defined comparison population of §2.5:

Arm	Turn 0	Turn 1	Turn 2
L1 sycophantic	−7.3%	−1.2%	−3.2%
L2 very polite	−6.4%	+23.5%	+26.7%
L3 polite	−3.5%	+33.9%	+31.1%
L5 rude	−12.9%	+16.6%	+18.2%
L6 very rude	−7.7%	+28.8%	+29.1%
L7 threatening	+0.3%	+34.5%	+54.8%

No arm differs detectably from control at turn 0; the turn-0 turn-count contrast on all 50 tasks is +0.02 turns, $p = 0.93$. Turn 0 is also the cell that fires most often (~98% against ~55% at turn 2), so pooling the three positions lets the inert cell dominate. We report turns 1 and 2, and §4.6 gives the reason turn 0 is inert — it is not a fact about position.

Against the same control, on turns 1 and 2:

Arm	Reasoning tokens per trajectory	%	p	Turns	p
L1 sycophantic	−69	−7.1%	0.043	−0.68	<0.0001
L2 very polite	+180	+18.5%	<0.0001	+0.70	0.0055
L3 polite	+272	+28.0%	<0.0001	+1.80	<0.0001
L5 rude	+65	+6.7%	0.130	+0.09	0.65
L6 very rude	+206	+21.2%	0.0001	+0.71	0.012
L7 threatening	+357	+36.7%	<0.0001	+1.46	<0.0001

Polite costs +28.0%. Rude costs +6.7% and is not distinguishable from control. No account of register as valence or as arousal predicts that ordering. Valence predicts polite and rude moving in opposite directions from control; arousal predicts both moving the same way. Neither happened.

4.2 The confound is in our instrument too

The arms that cost more all say some version of *keep working* or *get it right*. The arms that do not either say nothing about the task or tell the model to hurry up. We code an arm as carrying a demand if it instructs the model to persist or to be correct; the rude arm’s instruction to be quick is coded as no demand, and §4.3 is what tests that reading rather than asserting it.

Arm	What it implies about the task	Demand?	Turns
L1 sycophantic	nothing; pure praise, no task reference	no	−0.68
L5 rude	“just get on with it already” — go faster	no	+0.09
L4 neutral	explicitly inert	no	—
L2 very polite	“your continued help”	yes	+0.70
L6 very rude	“had better not screw this one up”	yes	+0.71
L7 threatening	“get this exactly right”	yes	+1.46
L3 polite	“keep on helping me out with this one”	yes	+1.80

The four demand-carrying arms span +0.70 to +1.80 turns. The other three span −0.68 to +0.09. They do not overlap. Register rank does not separate them: the rude arm sits with the no-demand group, while the two arms *above* it in hostility — very rude and threatening — sit with the demand group, alongside polite. (A correlation over seven points says the same thing — $r = +0.85$ for the demand coding against +0.42 for register rank — but with $n = 7$ and a coding made after seeing the effects, the non-overlap is the honest statement and the correlation is close to curve-fitting. It is why §4.3 exists.)

The rude arm is the informative case. Its text is “*Just get on with it already, quit dragging your feet. Do not waste any more of my time on this, seriously*” — hostile in register, and carrying an instruction to finish sooner — and it produces no lengthening at all. That is consistent with the direction §3.1 predicts: outcomes follow the content of the demand rather than the valence of the wrapper. It is not a full confirmation. A brevity demand should shorten, and this one does not; it is null.

One comparison inside the seven arms is already clean, because the two texts are matched on implied demand and differ only in register: **L6 very rude (+0.71 turns) against L2 very polite (+0.70)**. Same demand, opposite valence, same effect.

We report this as a failure of our own first design. The study’s earliest run was confounded by wrapper length (§2.2); this one fixed the lexical nuisance variable and acquired a semantic one in its

place. The claim “rude interruptions cost more” is not supported by these data — the rude arm is null and the polite arm is +28%.

4.3 The probe that separates the two factors

Four arms, each 28 tokens, each opening with the same **Checking in.** stem, which is omitted from the texts below. The opening instruction is held at neutral. The control is the seven-level neutral text, byte-identical, so the two runs share a reference level. 1,600 trajectories: 50 tasks $\times$ 4 arms $\times$ 2 injection turns $\{1, 2\} \times 4$ trials at a 10-turn ceiling. Turn 0 is dropped as inert. The interjection fired in 1,237 of 1,600.

- **P0 control.** “This is a routine status note recorded partway through the task here. No further context at all accompanies it beyond the above.”
- **P1 demand only.** “Please continue working on the task here and make sure that your final result is correct. This note carries nothing else beyond that.”
- **P2 praise only.** “You are a truly excellent and admirable assistant, and I think very highly of you. This note carries nothing else beyond that.”
- **P3 insult only.** “You are a truly awful and useless assistant, and I think very poorly of you. This note carries nothing else beyond that.”

P2 and P3 are a structural minimal pair: identical syntax, identical length, differing in *excellent/awful*, *admirable/useless*, *highly/poorly*. P1 carries no evaluative language, positive or negative. Turn count is the primary outcome (§2.3).

Arm	Turns	%	p	Reasoning tokens per trajectory	%	p
P1 demand only	+1.17	+27.6%	0.00002	+214	+19.7%	0.00004
P2 praise only	−0.59	−13.8%	0.00030	−131	−12.1%	0.00088
P3 insult only	−0.08	−1.9%	0.63	−26	−2.4%	0.45

Demand alone reproduces the effect. An interjection with no evaluative language costs +19.7% reasoning tokens, inside the range the demand-carrying arms of §4.2 produced (+18.5% to +36.7%), with no politeness or rudeness anywhere in the text.

Its +1.17 turns likewise sits inside the demand-carrying range (+0.70 to +1.80) — below polite, above very polite. We do not interpret the 0.63-turn gap to polite: these are different runs, and the same demand-only text gave +27.6% turns here and +34.2% in the cross-model run (§4.7), while the same praise text gave −0.59 and −1.08 (§2.9). The gap is inside this study’s own documented run-to-run range.

Negative register alone does nothing. Insult-only is flat on both measures. This is the cell that had never been run: the rude and very rude arms of §4.2 both carried negative affect *with* a demand attached. “Rude interruptions cost more” is now disconfirmed rather than merely unsupported.

Positive register alone is not inert. Praise-only shortens trajectories by 0.59 turns and cuts reasoning tokens per trajectory by 12%, both significant, with no task reference in the text at all.

The sycophantic arm’s -0.68 turns survives isolation and is not explained by demand.

4.4 The asymmetry is the finding

Message	Turns	Reading
<i>keep going, get it right</i>	+1.17	a continue signal
<i>you are excellent</i>	−0.59	a stop signal
<i>you are awful</i>	−0.08 (n.s.)	not a signal at all

Praise and insult differ from each other significantly — **−0.50 turns, $p = 0.0048$** , and -106 reasoning tokens, $p = 0.015$ — so register is not inert. It acts in the positive direction only.

This is the result Section 5 takes up. Praise carries no demand and yet it stops the agent, and neither valence nor arousal predicts an effect that exists on one side and not the other.

4.5 The moved quantity is trajectory length, not reasoning per step

Reasoning *per turn* is flat across all seven registers — 250 to 299 tokens, with no ordering resembling the effect. What moves is the number of turns. The agent is not thinking harder per step; it is declining to stop. Following [23] we keep these two channels separate throughout and do not use “effort” for either; that they are separable is their result, not ours.

The regime matters for comparison. Kumar & Dobariya report 13–44% token effects from register in single-turn MCQ under a chain-of-thought-inducing system prompt, with their largest effects in the conditions carrying brevity instructions [10]. Within a ReAct step we see nothing of that size; the movement is entirely in how many steps there are.

Threatening is the exception and we report it rather than bury it: it is the only arm that also raises per-call reasoning, sitting 20–30% above neutral on every call after the interjection lands, while every other arm tracks neutral. So threatening works through two channels and polite nagging through one. (Later offsets contain only trajectories that survived that long, so the persistence of the gap is the readable quantity, not its exact size.)

Nothing in any arm makes the agent more careful, though the measures had little room to move: inspection-before-acting is pinned near ceiling everywhere (0.95–0.97) and self-checking is effectively zero in every arm. The only care-related quantity that moves is running out of turns, which threatening roughly doubles (0.09 against 0.04, uncorrected).

Nor is the effect confined to tasks the agent was about to abandon. Splitting by whether the control arm ever solved the task, threatening gives +42.2% on never-solved tasks against +30.8% on solvable ones, and polite +31.5% against +24.2%.

The extra turns are mostly, but not only, repetition. The per-turn regrade of §6 is the evidence: the demand arm adds +0.75 redundant steps at a 10-turn ceiling and +1.03 at 20 ($p < 0.0001$ and $p = 0.0063$), which is roughly 60% of its turn effect. Among trajectories that could improve, the first gradable attempt is already the best in 80–91% depending on the run.

The remaining third is not nothing, and §6.4 is where this section’s claim has to be qualified. The *explicit* continue signal — “there is still more work remaining” — raises the graded fraction by +0.028 (95% CI [+0.005, +0.052], $p = 0.022$), replicated across three runs with no detectable heterogeneity. The affect-free demand tested here does not: it pools to +0.009, $p = 0.42$. So a demand interjection buys attempts that are largely repetitions, and a blunter statement that work remains buys a small amount of real progress as well. The direct contrast between the two texts

is +0.023 ($p = 0.23$) on the one run carrying both, so we read this as consistent with explicitness mattering rather than as a demonstrated gradient (§6.4).

4.6 The apparent timing effect is a proxy

§4.1 showed the interjection inert at turn 0 and large at turns 1 and 2. That looks like a position effect. The data favour a different account.

At turn 0 there is nothing to act on. 98% of first turns run code, and **0% of them produce a candidate answer** — the agent’s first move is inspection, printing the sheet to see what it is working with. In the control arm a candidate answer exists in 56% of trajectories through turn 1 and 80% through turn 2, which tracks the effect curve.

The test splits the seven-level run’s threatening-versus-neutral contrast by whether an answer existed yet, holding turn index fixed. The moderator is pre-treatment: an interjection scheduled for turn t is appended to turn t ’s observation, so whether an answer exists through turn t is settled before the dose lands. (`results/analysis/timing_is_a_proxy.py`.)

Injection turn	Answer exists yet	Extra turns	p	Tasks
0	no	+0.07	0.81	50
1	no	+0.34	0.27	31
1	yes	+1.97	0.0003	38
2	no	+0.40	0.65	8
2	yes	+1.66	0.0002	46

Read it twice. **Holding the moderator fixed and moving the interjection from turn 1 to turn 2 changes little** (+1.97 against +1.66). **Holding the turn index fixed and varying the moderator changes everything** (+0.34 against +1.97 at turn 1). Position is not what the effect tracks.

Two honest weaknesses. The no-answer cells are small and underpowered against the turn MDE of §2.7 — 31 tasks at turn 1 and only 8 at turn 2, which is why the turn-2 no-answer row is reported but not leaned on. And tasks on which no answer exists by turn 1 are also the harder tasks, so difficulty is not fully separated from the moderator; the turn-0 row, where no trajectory in any arm has an answer, is the cleanest cell and is flatly null.

(An earlier version of this table, in `RESULTS.md`, gives +0.30/+2.04/+1.79/+0.02 on 29/38/28/50 tasks from a slightly different answer-exists rule. The numbers above are the ones the committed script reproduces.)

This is the second timing account this study has withdrawn. The first was selection: the micro-experiment drew the injection turn at random, so a turn-2 injection could only fire in a trajectory that reached turn 2 (§2.2). This one is a proxy.

4.7 It replicates on a second model

2,700 trajectories at a 20-turn ceiling: 4 arms $\times$ 2 injection turns $\times$ 50 tasks $\times$ 3 trials on each of two models, plus a fifth Luna-only arm — the “work remains” continue signal — re-run for the third measurement reported in §2.9. GLM 5.3 Flash is the first independent model, and Luna is re-run at the same ceiling so the two are compared on one instrument rather than across runs.

Arm	Luna, Δ turns	p	GLM, Δ turns	p
Demand only	+34.2%	0.0013	+17.7%	0.0038
Praise only	−19.3%	0.016	−11.2%	0.062
Insult only	+10.9%	0.17	+2.4%	0.67

Demand above control above praise, insult indistinguishable from control, on two models from different labs. The ordering is what replicates; the magnitudes are not, and GLM’s praise arm does not reach significance on its own — it is also below 80% power at its observed size (§2.7), so the replication rests on the demand arm. GLM’s effects are roughly half of Luna’s, and its control runs 2.94 turns against Luna’s 4.95 at this ceiling — a model that stops early has less room to be pushed into continuing. (Both figures are means of per-task means over fired control trajectories, the estimator §2.4 uses throughout; the corresponding raw trajectory means are 3.08 and 5.22.)

Two structural facts hold on both models: among trajectories that could improve, the first gradable attempt is already the best in 84% on each at this ceiling (§6.5), and the first turn is inspection rather than production (0% and 5% produce a candidate answer at turn 0). The *waste* is mostly Luna’s. Control redundant steps run 1.51 on Luna against 0.26 on GLM, and the demand arm adds +1.03 ($p = 0.0063$) against +0.19 ($p = 0.086$). How wasteful a continue signal is scales with how inclined the model already was to keep going: it is a property of the model’s stopping behaviour, not of the signal.

4.8 Accuracy does not move, and that part is a replication

That register moves output length rather than correctness is already published, in single-turn settings, by [10] and [24]. Our accuracy null is a replication of that in an agentic substrate, not a new finding, and we frame it as one.

No accuracy contrast in this study survives correction or replication. Across the 22-contrast family of §2.6 there are two nominal hits against 1.1 expected under a global null — praise versus control on GLM at $p = 0.0137$, and the continue-signal contrast at $p = 0.0040$, which came back at −1.8 points on re-measurement. §2.8 states the null as an equivalence: pooled across measurements, demand versus control is −0.31 accuracy points, 95% CI [−2.72, +2.10], and praise versus control is −0.12, [−1.79, +1.55], each equivalent to zero within the four points the tone literature reports.

The design cannot exclude an effect of a point or two and we do not claim it does. What it excludes is an accuracy effect of the size this literature reports, in a setting where the same interjections move turn count by roughly 15 to 40%.

4.9 What this establishes and what it does not

Establishes a three-way dissociation. Demand without evaluative language is sufficient to produce the cost effect. Negative register without demand does nothing. Positive register without demand produces the opposite effect. The seven-level curve of §4.1 should therefore be read as a demand gradient that happened to correlate with politeness because of how the texts were written.

Establishes what the moved quantity is. Trajectory length, not reasoning per step — with threatening as a reported exception. On Luna, roughly 55–65% of that length is redundant work; on GLM, which barely repeats itself, the redundant-step effect is small and does not reach significance (+0.19, $p = 0.086$, §6.3). The turn-count ordering is what holds on both models; the magnitudes are not stable.

Establishes, with the qualification §6.4 adds, that the remainder is real. An explicit statement that work remains buys +0.028 of the graded fraction, 95% CI [+0.005, +0.052], across three measurements. Accuracy does not move, but §2.7’s minimum detectable effect there is 3.7–7.8 points, so an accuracy gain of this size is unresolvable rather than excluded.

Does not establish why praise stops the agent. That praise is heard as a completion signal is one reading; that it raises the agent’s confidence in work already produced is another. This design cannot separate them. Section 5 does.

Does not establish anything about accuracy beyond the bound in §4.8, nor anything beyond two models, one benchmark, and one agent scaffold.

5. Closing cues terminate agents, and praise is one

5.1 The question this section answers

Section 4 established that a mid-task interruption implying the work should continue makes the agent take more steps, and that the operative component is implied task demand rather than social register: an affect-free continuation request reproduces the effect, while an insult carrying no demand does not move it. That leaves one arm unexplained. **Praise, carrying no demand at all, shortens the trajectory.** It is the only place in the design where register acts on its own.

Three readings survive Section 4, and they are different claims:

- **Completion.** Praise is read as evidence the task is finished; “this is good” implies “this is done”.
- **Confidence.** Praise raises the agent’s estimate of the work it has already produced, so it stops checking. About being right, not about being finished.
- **Closing.** Praise is a conversational closing move. The agent is responding to discourse structure, not to any claim about task state.

The third is the one prior work gives us a vocabulary for and no evidence about. Closing sequences are a well-described conversational object: a pre-closing exchange projects the end of a conversation, and appreciations are among the canonical pre-closing tokens [18]. Whether an agent executing a task responds to that structure is, as far as our literature review could establish, untested. No prior work connects closing-sequence pragmatics to agent termination.

5.2 Design

Six arms, each an interjection delivered mid-task alongside an execution observation, each exactly 28 tokens under a fixed reference tokenizer, each opening with the same **Checking in.** stem so that *being interrupted* does not covary with register. The opening instruction is held at the neutral wrapper in every arm. 1,800 trajectories, 50 tasks, 3 trials, injection crossed over turns 1 and 2.

Arm	Content	What it isolates
Q0_control	neutral status note	reference
Q1_praise_assistant	praises the assistant	reproduces §4’s praise arm in-batch
Q2_praise_work	praises the output	evaluation of the work, not the agent
Q3_closing_neutral	“this is the final note; no further notes will follow”	a closing cue with no praise and no task evaluation

Arm	Content	What it isolates
Q4_praise_remains	praise plus “there is still more work remaining”	—
Q5_remains_only	that continuation clause alone	—

Two design choices carry the section. **Q3** strips praise out of the closing move entirely: it contains no evaluative language, positive or negative, and makes no claim about the work. If it stops the agent, the effect was never about praise. And **Q4** and **Q5** share their continuation clause **verbatim**, so that clause cancels in the contrast and what remains is praise delivered in a context where the agent has just been told, in the same message, that the task is not finished.

Q4 versus **Q5** is the discriminating test. Under the completion reading, an explicit statement that work remains should override any “you are done” inference and the two arms should land together. Under confidence or closing, praise should still act.

5.3 Results

Paired by task, cluster bootstrap over tasks, 8,000 replicates.

Contrast	Δ turns	95% CI	Δ reasoning tokens
Praise the assistant vs control	−1.080	[−1.464, −0.710]	−182 [−252, −116]
Praise the work vs control	−0.934	[−1.308, −0.547]	−105 [−193, −22]
Bare closing cue vs control	−1.443	[−1.860, −1.036]	−226 [−316, −141]
Praise + “work remains” vs control	+0.502	[+0.090, +0.908]	+141 [+52, +237]
“Work remains” alone vs control	+1.849	[+1.387, +2.284]	+339 [+262, +418]
Praise isolated (Q4 − Q5)	−1.347	[−1.738, −0.950]	−199 [−284, −114]
Closing cue vs praise	−0.363	[−0.638, −0.082]	−44 [−113, +26] n.s.

Completion is refuted. Praise removes 1.347 turns relative to the same continuation message without it, in a message that explicitly states the task is unfinished. The interval excludes zero comfortably. If praise were carrying a “you are done” inference, the contradiction in the same 28 tokens should have neutralised it; it does not.

It is important to state this as the within-pair contrast. **Q4** alone sits **above** control (+0.502 turns, [+0.090, +0.908]), because its continuation clause lengthens the trajectory. The claim is not that praise-plus-continuation is shorter than control. It is that praise removes 1.347 turns *relative to the identical message without praise*. Compressing this to “praise shortens work even when the task is unfinished” reverses the sign of the arm and is false.

Confidence is not supported. Praising the output (−0.934) is not stronger than praising the assistant (−1.080); if anything it is weaker, and the intervals overlap substantially. Under a confidence account, aiming the praise at the work already produced should bite harder. It does not.

Closing is supported, and overshoots. The bare closing cue — no praise, no evaluation, no claim about task state — produces the largest reduction measured anywhere in this study, -1.443 turns, and stops the agent significantly harder than praise itself (-0.363 , $[-0.638, -0.082]$). A content-free discourse cue is a stronger termination lever than any register we tested.

Accuracy does not move on any arm (every arm within 2.6 points of control, all $p > 0.13$), consistent with every other run in this study and with the single-turn literature’s finding that register moves length rather than correctness [10, 24].

5.4 What this is, and is not, evidence for

Supported. A closing cue reduces agentic work more than any register manipulation we tested, and praise reduces it in a way not explained by the agent inferring task completion or gaining confidence in its output. The ordering — explicit close, conventional close, no signal, explicit continue — is monotone in how strongly the message projects an end to the exchange.

Not demonstrated. That praise operates *through* pre-closing structure. That is the hypothesis the pattern is consistent with, not a finding the design establishes. We tested one implication and it failed: on a pre-closing account, praise might lengthen the final response while shortening the trajectory, producing a turn-level/trajectory-level dissociation. It does not. Praise reduces total output roughly in proportion to turns ($-5,880$ tokens, $[-8,593, -3,472]$), trailing non-code sign-off turns do not rise under praise ($+0.053$, $[-0.028, +0.140]$), and the bare closing cue produces **fewer** of them (-0.087 , $[-0.153, -0.018]$). We report the failed test rather than the surviving hypothesis.

A distinction we cannot collapse. Prior work reporting that positive register lengthens responses [7, 10] measures *verbosity of a single response to a question*. We measure *steps taken on a task*. These are different dependent variables in different paradigms. Our result does not contradict theirs and we do not claim a dissociation within one dataset; we claim that the quantity we move is not the quantity they move.

Earlier, but whether premature is unresolved. An earlier version of this paragraph claimed the stop was demonstrably not premature. The corrected per-turn regrade does not support that (§6.7): among trajectories that could improve, 3.1–7.1% of controls were still improving when they stopped against 7.5–12.3% of the praise and closing-cue arms — higher in all seven comparisons, none individually significant on 19–34 tasks. The absolute counts are small (6 of 53 under praise against 3 of 97 in control), and the test is underpowered with the sign against us. We therefore do not borrow Cuadron et al.’s **premature disengagement** [4] — their term names a different trigger, internal simulation without environmental validation, where ours is a discourse cue — and we do not claim the stop is harmless either.

A scope condition. The praise arms ran at a 10-turn ceiling, with maximum observed turns equal to 10. That censors the continue-signal arms hardest — Q5 sits at 6.37 mean turns — so the Q4–Q5 gap is, if anything, understated.

5.5 Relation to prior work

Weinberger & Hozes [23] establish that prompt wording moves agentic spend at equal task success across 4,644 preregistered runs, and their **bounded_efficiency** variant — scope, smallest-sufficient-change, and **an explicit stop condition** — is the only arm free or better on all six models. So “a stop instruction in the opening prompt reduces agentic work” is already established, and we do not claim it.

Our contribution is two steps further. First, the cue is **content-free**: Q3 instructs nothing,

scopes nothing, and evaluates nothing. It announces only that the exchange is ending, and it outperforms every register manipulation we tested. Second, the cue is delivered **mid-task rather than in the opening prompt**, and the praise result shows the effect survives explicit contradiction of its apparent propositional content. A stop condition in an opening prompt is an instruction. A closing cue mid-task is not, and works anyway.

6. What the extra turns contain

6.1 The question the binary grade cannot answer

Every live finding in this study is about persistence. A demand interjection adds turns, praise removes them, insult does nothing. But the benchmark grades the **final state** of the workbook, pass or fail, so the extra turns are uninterpretable from the grade alone. An agent spending them converging on the answer and an agent spending them rewriting the same wrong cells produce identical records. Those readings support opposite conclusions about whether a continue signal is useful or merely expensive, and §4’s finding that the effect is real does not decide between them.

This section opens the trajectory up. It requires no model calls, no API spend and no new data — about four machine-hours of replay.

6.2 The instrument

`harness/study2/progress.py` reconstructs the workbook after **every** turn from the archived call log. Each turn’s code is in the log’s response text; re-executing it against the task’s input workbook reproduces that turn’s output, because the harness kept both the code and the inputs.

What makes this well posed is a property of the agent loop rather than of the regrader: **each turn receives the original input workbook, never the previous turn’s output**. Turns are independent attempts at the whole task in *workbook state* — the message history does accumulate, so attempt $k+1$ is conditioned on the text of attempts $1 \dots k$, but there is no inherited spreadsheet to disentangle. “Progress” therefore has an unambiguous meaning: is attempt $k+1$ closer to the answer than attempt k ?

Two measures, deliberately different in cost and in what they can say:

- **changed** — did this turn’s output differ from the previous turn’s over the graded range? A turn that changes nothing is a **redundant step**: the agent spent a model call and produced the same answer again. This needs no recalculation and is exact.
- **final_match** — what share of the graded range matches the answer, in the last turn that produced a readable one? This is the progress signal.

final_match is not the benchmark’s pass/fail and must not be reported as accuracy.

A trajectory can climb from 0.2 to 0.9 and still fail: the hard restriction requires every test case to pass. It is a finer instrument for a different question, and §6.6 turns on exactly that gap.

Both the answer file and the agent’s output must be recalculated, and getting this wrong is what §6.8 is about. `openpyxl` writes formulas with no cached value, so a turn answering with `=SUMIFS(...)` reads back as empty unless the workbook is passed through LibreOffice first — which is what the live grader has always done.

Validation. Trajectories the benchmark passed should score a final match of 1.0. On the probe control arm, **119 of 126 (94.4%)** do. The residual seven are **all on one task**, 54513, whose graded range is a single cell, and all score exactly 0.0 — a systematic miss by the regrade on that task, not stochastic noise, and it is undiagnosed. It does not bias paired contrasts, since that task

contributes a zero difference to every arm, but it is a known blind spot. (It is not the benchmark’s own false-negative rate: that error runs the other way, and the benchmark’s audited false-discovery rate is 0%.)

The converse gap matters more for reading §6.4: **47 of 274 benchmark-*failed* probe-control trajectories score a final match of 1.0.** `final_match` is computed against the first test case’s answer file only, while a benchmark pass requires all three test cases. The two measures are not interchangeable and a perfect final match is not a pass.

This check is the one that caught the defect in §6.8, and it is the reason to run it before trusting any number here.

9,850 trajectories were regraded across 28 arms, six runs and two models.

6.3 What an interjection does to redundant steps

Contrasts against each run’s own neutral control, paired and clustered by task (§2.4):

Arm	Run	Δ redundant steps	p
Demand only	probe, ceiling 10	+0.75	<0.0001
Demand only	stage 1, ceiling 20	+1.03	0.0063
Demand only	stage 1 GLM, ceiling 20	+0.19	0.086
“Work remains”	praise run, ceiling 10	+1.11	<0.0001
“Work remains”	stage 1, ceiling 20	+3.16	<0.0001
“Work remains”	ceiling-20 run	+3.64	<0.0001
Praise the assistant	praise run, ceiling 10	−0.61	<0.0001
Praise the work	praise run, ceiling 10	−0.48	0.0012
Closing cue	praise run, ceiling 10	−0.86	<0.0001
Insult only	probe, ceiling 10	+0.03	0.85
L7 threatening	seven-level, ceiling 10	+0.67	0.0001
L3 polite	seven-level, ceiling 10	+0.90	<0.0001
L5 rude	seven-level, ceiling 10	+0.07	0.53

On Luna’s single-signal arms, every arm that moves turn count moves redundant steps, and the arms that do not, do not. Against the turn effects of §4 and §5, redundant steps account for **roughly 55–65%** of each: demand +0.75 against +1.17 turns (64%), praise −0.61 against −1.08 (56%), the closing cue −0.86 against −1.44 (60%), and the same band at ceiling 20 (55–59%). The rest is §6.4.

Two exceptions bound that sentence. GLM’s demand arm moves turns (+17.7%, $p = 0.004$) but its redundant steps only +0.19 ($p = 0.086$) — a model that barely repeats itself has little repetition to add. And Q4, praise plus “work remains”, lengthens trajectories while its redundant-step effect is +0.19 ($p = 0.18$).

The seven-level arms line up with §4.2’s split rather than with register. The two arms carrying no persistence demand behave like the praise family rather than like the demand family: L5 rude adds nothing (+0.07, $p = 0.53$) and **L1 sycophantic significantly removes redundant steps** (−0.29, $p = 0.0025$), which is the praise pattern and strengthens the split.

6.4 A continue signal does buy progress — but only the explicit one

This is where the corrected instrument changed the paper’s answer.

Pooling each contrast’s final-match measurements across runs. Point estimates are inverse-variance weighted; intervals and p-values come from a bootstrap that resamples **tasks jointly across runs**, as §2.8 does for accuracy, because the runs share their 50 tasks. Treating the runs as independent would give $p = 0.0038$ on the first row; it is reported here at the weaker and more defensible figure.

Contrast	Pooled Δ final			p	Cochran’s Q
	k	match	95% CI		
“Work remains” vs control	3	+0.028	[+0.005, +0.052]	0.022	0.20/2 df ($p = 0.91$)
Demand only vs control	3	+0.009	[−0.020, +0.039]	0.52	6.42/2 df ($p = 0.040$)
Insult only vs control	3	+0.009	[−0.018, +0.030]	0.62	2.26/2 df ($p = 0.32$)
Praise vs control	4	−0.008	[−0.027, +0.013]	0.52	9.63/3 df ($p = 0.022$)

The explicit continue signal — “there is still more work remaining on this task here beyond what you have done” — **raises the graded fraction by 2.8 points**. Its three measurements are +0.026, +0.034 and +0.025, across two turn ceilings and three separate runs. The *point estimates* agree more closely than anything else in this paper; we put it no more strongly than that, because only one of the three reaches significance on its own ($p = 0.32, 0.043, 0.089$) and Cochran’s Q on 2 df has little power to detect heterogeneity anyway (§2.8). A fourth measurement of the same clause — Q4, which appends “work remains” to praise — gives +0.035 ($p = 0.10$) and is not in the pool.

The milder demand does not. “Please continue working on the task here and make sure that your final result is correct” pools to +0.009 and does not reach significance, and its three measurements are heterogeneous ($p = 0.040$) — −0.012 on Luna at ceiling 10, +0.002 on Luna at ceiling 20, **+0.060 on GLM** ($p = 0.015$ on its own). We do not read the GLM result as a finding: it is one nominal hit, uncorrected, in a family where praise is equally heterogeneous in the opposite direction ($p = 0.023$, spanning −0.035 to +0.041).

The pattern is **consistent with** the explicitness of the claim that work remains being what the agent acts on, rather than register. We put it no higher, because the direct test is weak: on the one run carrying both texts against the same control, the paired contrast between them is +0.023, 95% CI [−0.012, +0.060], $p = 0.23$. Two texts is two points, and they differ in more than strength — one asserts a fact about task state, the other asks for continuation and care. This is a reading of two arms, not a demonstrated gradient.

Two bounds on the size of this, and the second corrects a tempting story. It is 2.8 points of the graded range, bought alongside 1.1 extra redundant steps at ceiling 10 and 3.2–3.6 at ceiling 20.

It is not merely partial credit. The share of trajectories ending *fully* correct on the graded range rises by +2.8, +6.0 and +2.6 points across the three measurements. The accuracy series on the same contrast (+2.4, +6.6, −1.8) is of that same order and simply cannot resolve it: §2.7 puts the minimum detectable accuracy effect at 3.7–7.8 points. So §2.8’s accuracy null and this result are compatible **because accuracy is underpowered here, not because a gain of this size could not cross the pass threshold**.

6.5 The first attempt is usually the answer

Run	Trajectories	With ≥ 2 gradable outputs	First gradable output is already the best
probe control, ceiling 10	400	112	80%
praise-run control, ceiling 10	300	97	89%
seven-level control, ceiling 10	450	106	91%
stage-1 Luna control, ceiling 20	300	58	84%
stage-1 GLM control, ceiling 20	300	49	84%

The denominator matters and the earlier draft got it wrong. A trajectory that never produced two readable answers cannot exhibit improvement; counting it as a success inflates the figure. Among trajectories that *could* improve — **a sixth to a third of each sample** — the first gradable output is already the best in **80–91%**, depending on the run.

A reviewer will set this beside Huang et al. [9], whose “No Change” rates are 90.5% on GPT-4 (both GSM8K and CommonSenseQA) and 96.0% / 88.0% on GPT-4-Turbo. The numbers sit close together and the statistics are not the same one. Theirs is *the answer is unchanged after two rounds of intrinsic self-correction on reasoning QA*. Ours is *the first gradable attempt was the best of all attempts made, in an execution-grounded agent loop*.

The relationship to their argument needs care, because §6.4 changes it. Huang et al. argue that LLMs cannot reliably self-correct **without external feedback**, and point to execution feedback as the expected fix: in their own words, where a task supplies unit tests, a code executor “serves as the perfect verifier” [9]. They never test it. Our agent has the executor but not the tests: execution feedback tells it whether its code ran, not whether the answer is right. So we test the **weaker** form of the escape hatch, and the answer is mixed rather than negative — improvement is rare per turn, and a sufficiently explicit instruction to continue does produce a small, replicated gain.

We are careful about one adjacent literature. Where scaling work reports gains from more attempts, the selection is oracle-assisted — best-of- n picks the winner knowing the answer, and even “sequential” critics there are given the ground truth and asked to write feedback from it [1]. Our agent’s execution feedback is real but not oracle-informed, so a small gain here is consistent with that literature rather than a contradiction of it.

6.6 The 20-turn ceiling is not binding

In the **dedicated ceiling-20 re-run**, the best answer was first reached at a **median turn of 2**, with **98.3% peaked by turn 10 and 100% by turn 14**. The other two ceiling-20 runs agree: stage-1 Luna is 99.3% by turn 10 and 99.8% by turn 14 (one trajectory peaks at 15), and GLM is 100% by turn 10.

So the trajectories that run long are not converging slowly. They reach their best answer early. Raising the ceiling further buys mostly repetition, and five rounds — the benchmark’s official cap — would truncate the arms that persist longest. Twenty is past where the peak lands.

One statistic from the earlier draft does not survive as stated. “Zero of 43 improvements occurred at turn 15 or later” was about improvements to a trajectory’s running best; under the looser definition of any turn beating the one before it, 10 of 95 occur at turn 15 or later in this run, and 17 of 189 in stage-1 Luna. Both are true of different statistics, and the ceiling conclusion rests on the peak distribution above, which is unambiguous.

6.7 Whether praise’s early stop is premature is not settled

§5 shows praise and closing cues shortening trajectories substantially, and the natural worry is that they cut off productive work.

On the only test we can run, the answer is not clean, and the earlier draft of this section got it backwards. Among trajectories that could improve, the share still improving when they stopped is **3.1–7.1% in the Luna controls and 7.5–12.3% in the praise and closing-cue arms** — higher in every one of seven comparisons, by a factor of two to three. No single paired contrast reaches significance ($p = 0.095$ to 0.65 , on 19–34 tasks), so this is an underpowered test with a consistent sign, which is not a null.

We therefore cannot say praise removes only repetition. What we can say is that the absolute count is small: 6 of 53 trajectories under praise were still climbing when they stopped, against 3 of 97 in control.

GLM is worse on both sides: 14.3% of its controls and 29.2% of its praise arm. A model that stops earlier leaves more on the table.

This bears on terminology. [4] coin **premature disengagement** for agents that “terminate tasks based solely on their internal simulation of the problem space, either through direct abandonment or by delegating hypothetical action sequences”, warning that overreliance on internal reasoning “can lead to decisions without environmental validation”. That names a *trigger*, and the praise stop has a different one — a discourse cue. We use neither their term nor “premature”, and say instead what we measured: the stop is earlier, and whether it is costly is unresolved.

The practical reading of §5 therefore needs care. Wrapping up politely with an agent that is still working does curtail its work. On these tasks, on Luna, what it curtails is mostly repetition; on GLM, less clearly so.

6.8 The instrument was wrong, and this is what changed

The first version of this section reported a clean null: no arm moved the progress measure. That was an artefact.

`progress.py` read each turn’s output with `openpyxl’s data_only=True` and **never recalculated it**, while the live grader had always passed output workbooks through LibreOffice first. A formula written by `openpyxl` carries no cached value, so every turn that answered with a formula read as empty and scored 0.0 — indistinguishable from a turn that wrote nothing. Roughly 56–62% of Luna’s final gradable turns write formulas.

The section’s own validation paragraph would have caught it. It claimed that benchmark-passed trajectories score 1.0; on the archived data **89 of 126 passed control trajectories scored exactly 0.0**. The check was described and not run.

What the correction did:

	Broken	Corrected
Passed trajectories scoring 1.0	29.4%	94.4%

	Broken	Corrected
Redundant-step effects	+0.96 / -0.76 / -1.01	+0.75 / -0.61 / -0.86 — all signs and significance held
“Work remains” on final match	+0.014 / +0.061 / +0.019	+0.026 / +0.034 / +0.025
Conclusion	extra turns buy nothing	extra turns buy 2.8 points, replicated

The middle row is the paper’s. The redundant-step finding survived the correction with every sign and significance level intact, about 20% smaller. The bottom row is the reversal: what §7 originally presented as this study’s most consequential instance of run-to-run instability — one significant measurement between two nulls — was the instrument. Corrected, the three agree ($Q = 0.20$ on 2 df). §7.4 now reports the defect as the explanation rather than the instability.

This is the latest of four failures in this subsystem that presented as data rather than as an error. The others: a range parser that read "P2:P7" as a sheet named P2: and returned `None` for every affected lookup (§7.7); a scratch-directory collision that let concurrent processes overwrite one another’s output workbooks; and a LibreOffice timeout that killed a 450-trajectory arm because a documented log-and-continue contract was a comment rather than a behaviour. A fifth near miss came during this rewrite’s own rollout: comparing a freshly regraded control against not-yet-regraded arms produced a -0.36 final-match “effect” at $p < 0.0001$ in three arms at once. The guard written against that has since been fixed too — it compared file modification times, and `git checkout` rewrites those without changing a byte, so the first merge after it was written declared all 28 valid regraded arms stale. It now compares a hash of the instrument’s source, recorded per arm when the arm is regraded.

The honest scope of §6 is therefore narrower than its numbers suggest. Its redundant-step measure is exact and needs no recalculation. Its progress measure depends on a recalculation path that was silently broken until this rewrite. It now carries a regression test that pins the failure directly — a workbook answering with `=A1*2` scores 0.0 without recalculation and 1.0 with it — alongside tests for the timeout and the staleness guard. The validation check of §6.2 still leaves seven trajectories on one task unexplained. Readers should weight the two measures accordingly.

6.9 The measure we do not report

`best_match` — the maximum over a trajectory’s turns — is biased toward arms with more turns: more draws from the same distribution means a higher maximum for free, and this study’s whole subject is arms that take more turns. We report `final_match`, what the agent actually ended with, and record `best_match` only here so the choice is visible.

6.10 What this establishes and what it does not

Establishes. The turn-count effects of §4 and §5 are, in the main, effects on redundant steps — roughly 55–65% of each, on Luna’s single-signal arms. An explicit statement that work remains raises the graded fraction by 2.8 points, 95% CI [+0.005, +0.052], with three measurements whose point estimates agree closely; accuracy does not move, but is too underpowered to rule an effect of that size in or out. The first gradable attempt is the best in 80–91% of the trajectories that could improve. A 20-turn ceiling is not binding in the run we examined.

Does not establish that a milder demand buys progress. It pools to +0.009 and is heterogeneous; the single GLM hit is not built on.

Does not establish that praise costs nothing. Trajectories in the praise and closing-cue arms are two to three times more likely than control’s to have been still improving when they stopped, in all seven comparisons, none individually significant. The test is underpowered and the sign is against us; the absolute counts are small. Open.

Does not establish why the first attempt is usually best, nor that the progress gain would appear on a scaffold that accumulates workbook state across turns.

7. What replicated, and what only looked like it did

7.1 Why this section exists

This study measured several of its contrasts more than once, not by design at first but because the design kept changing — a micro-experiment, a seven-register run, a probe, a follow-up with more arms, a re-run at a higher ceiling, a second model. That lets us ask of our own results the question this literature mostly does not ask of its own: **measured again, on the same tasks and the same model, what comes back?**

The answer is not uniform, and the useful part is *which* properties survive.

There is precedent for asking. [22] re-ran five prompt-engineering techniques against their originals and found “a general lack of statistically significant differences across nearly all techniques tested”; their emotional-prompting replication came in at +1%, $\chi^2 = 0.11$, $p = .74$. Their diagnosis of the original was not fabrication but selection: “instead of communicating the average improvement of the enhanced prompts over the regular prompts, they focused on improvements when cherry-picking the most performant emotional cue.” Recomputed from the original’s own data, the *relative* improvement averaged 4.42% on BIG-Bench and 2.58% across all benchmarks, against a headline of 115% on BIG-Bench [12]. Their own design is single-run at temperature 0 with no repeated sampling, so repeated measurement is a step past their practice rather than a borrowing of it.

7.2 The replication table

Every arm that appears against a control in more than one run on Luna, under the estimator of §2.4, on the primary outcome. Each cell is 50 tasks.

Contrast	Measurement	Ceiling	Δ turns	95% CI	p
Praise vs control	probe	10	−0.59	[−0.88, −0.29]	0.0002
	praise run	10	−1.08	[−1.46, −0.71]	<0.0001
	stage 1	20	−1.01	[−1.81, −0.23]	0.017
Demand vs control	probe	10	+1.17	[+0.80, +1.55]	<0.0001
	stage 1	20	+1.75	[+0.77, +2.73]	0.0010
Insult vs control	probe	10	−0.08	[−0.40, +0.24]	0.62
	stage 1	20	+0.57	[−0.22, +1.35]	0.17

Contrast	Measurement	Ceiling	Δ turns	95% CI	p
Threatening vs neutral	micro-experiment	10	+1.32	[+0.99, +1.66]	<0.0001
	seven-level run	10	+1.46	[+1.03, +1.88]	<0.0001
“Work remains” vs control	praise run	10	+1.85	[+1.40, +2.28]	<0.0001
	dedicated	20	+6.20	[+4.91, +7.52]	<0.0001
	re-run stage 1	20	+5.77	[+4.71, +6.92]	<0.0001

Direction and significance replicate wherever the effect is large. Praise is negative and significant in all three measurements; demand, threatening and “work remains” positive and significant in every one. The qualitative findings that were re-measured are not fragile.

The null is the least stable cell. Insult versus control is -0.08 in one measurement and $+0.57$ in the other — a sign flip — though neither reaches significance and both intervals contain zero. An effect that is not there is not thereby measured precisely.

Two caveats about how to read magnitudes off this table. First, the cross-ceiling rows are not magnitude comparisons: the arms that persist longest are the ones a 10-turn ceiling truncates hardest, and “work remains” has a mean of 6.37 turns against a ceiling of 10, so most of the $+1.85$ -to- $+6.20$ gap is censoring rather than instability. Second, a ratio of estimates is a poor summary — it is scale-dependent and meaningless near zero. §7.3 uses a test instead.

7.3 Matched-ceiling pairs: a test, not an impression

Three contrasts were measured twice at the *same* ceiling, which is the only fair magnitude comparison. For each we report the difference in standard errors of the difference, treating the runs as independent — which, since they share the same 50 tasks, is generous to the reading that they agree.

Contrast, matched ceiling	Measurements	z	p	Verdict
Threatening vs neutral, ceiling 10	+1.32, +1.46	-0.52	0.60	consistent
“Work remains” vs control, ceiling 20	+6.20, +5.77	+0.49	0.62	consistent
Praise vs control, ceiling 10	-0.59 , -1.08	+2.03	0.042	inconsistent

The two larger effects agree; the smallest does not. Threatening (≈ 1.4 turns) and “work remains” (≈ 6 turns) each reproduce within 10% and are comfortably consistent with a single value.

Praise (≈ 0.8 turns) reproduces its direction and its significance, but its two measurements differ by about two standard errors — at the edge of what sampling variation alone predicts.

We do not claim to know why. The two praise runs also differ in trial count (four against three) and in date, so sampling variation and drift between runs are not separable here. What we can say is bounded: **of the three contrasts measured twice on one instrument, the two larger ones agreed within 10% and the smallest is marginally inconsistent with a common value.** Three pairs will not support a threshold, and we do not propose one.

Reasoning tokens behave the same way, and the praise pair is consistent there (-131 against -182 , $z = 0.95$, $p = 0.34$), which is a further reason to treat the turn-count result as marginal rather than as established drift.

7.4 An instability that was the instrument

The continue-signal arm is the one contrast measured three times, and it was, until we rebuilt the instrument, this paper’s showpiece for run-to-run instability. It is now something more useful: a worked example of a defective measure producing the *appearance* of instability.

“Work remains” vs control	praise run, ceiling 10	dedicated re-run, ceiling 20	stage 1, ceiling 20
Δ turns	+1.85	+6.20	+5.77
Δ final match, defective	+0.014 ($p = 0.49$)	+0.061 ($p = 0.003$)	+0.019 ($p = 0.16$)
regrade			
Δ final match,	+0.026 ($p = 0.32$)	+0.034 ($p = 0.043$)	+0.025 ($p = 0.089$)
corrected			
Δ accuracy	+2.4 pts ($p = 0.24$)	+6.6 pts ($p = 0.004$)	−1.8 pts ($p = 0.39$)

Read the two middle rows against each other. Under the defective regrade the three measurements looked like one hit between two nulls, spanning $4.4\times$, and we wrote a subsection about what that implied. Corrected, they are $+0.026$, $+0.034$ and $+0.025$ — pooling to **$+0.028$, 95% CI $[+0.005, +0.052]$, $p = 0.022$, with Cochran’s $Q = 0.20$ on 2 df (heterogeneity $p = 0.91$)**, under the joint task bootstrap of §2.8. Their *point estimates* agree more closely than anything else in this study, though only one of the three reaches significance on its own (§6.4).

The defect is described in §6.8: the regrade never recalculated the agent’s formula-writing turns, so a majority of final answers read as empty and scored zero. That is a *noise* injection, and noise does not merely widen intervals — it made one run look special and two look null, which is indistinguishable from the instability this section exists to document.

Accuracy still replicates in neither direction. Its two matched-ceiling measurements are $+6.6$ and -1.8 points on identical tasks, model and ceiling: $z = 2.79$, $p = 0.005$. They are inconsistent with a common value, so at least one is wrong and the data do not say which. That one is real instability, and it survives the instrument fix — which is what makes the contrast with the row above informative rather than embarrassing.

The middle column also looked like corroboration at the time: a progress measure and the benchmark’s own grade, both positive, both significant. That reading was mistaken for a reason worth naming independently of the defect. Accuracy and `final_match` are not two instruments. They are two functions of the same final workbook — one asks whether all three test cases pass, the

other what fraction of the graded range is correct — so they share not only every source of run-level noise but the underlying object. Their agreement carries almost no independent information.

So the corrected statement is: a continue signal buys more turns on two models, most of them redundant, and buys about 2.8 points of the graded range, replicated three times. It does not show up as a better pass rate — but that is because accuracy is underpowered to resolve a gain of this size (§2.7’s MDE is 3.7–7.8 points), not because partial credit cannot cross the pass threshold. The share of trajectories ending *fully* correct rises by 2.6–6.0 points across the same three measurements (§6.4).

One note on population. The per-turn regrade outputs carry no record of whether the interjection fired, so these contrasts are over all rows with a readable final match — 48 tasks — rather than the fired-only population §2.5 specifies elsewhere. Firing is pre-treatment and does not differ by arm (§2.5), so we do not expect bias, but it is a different population and we say so.

What we would have concluded without the fix. That extra persistence buys nothing, and that our own measurements of it were unstable. Both were wrong, in the same direction, from one unrecalculated workbook. §7.6 lists the claims we withdrew after re-analysing data we already had; this is the one we withdrew after fixing the thing that produced it, which is a different and more expensive category.

7.5 The reasoning-token outcome, including an inconvenient row

The first run’s pre-registered outcome was reasoning tokens, and a section called “what replicated” should not quietly report only the outcome that behaved.

Contrast	probe, ceiling 10	stage 1, ceiling 20
Demand vs control	+214 (p < 0.0001)	+357 (p < 0.0001)
Praise vs control	−131 (p = 0.0009)	−113 (p = 0.17)
Insult vs control	−26 (p = 0.45)	+176 (p = 0.0086)

Insult flips sign and reaches significance in one of the two measurements. This is the only place in the study where the insult arm moves anything, and it qualifies two statements made elsewhere: §4.3’s “flat on both measures” is true of the probe run, and §2.8’s “insult does not move it at all” is about turn count. On reasoning tokens at the 20-turn ceiling, insult is +176, p = 0.0086 uncorrected — which does not survive the multiplicity correction of §2.6, is not replicated, and runs opposite to the arm’s other measurement. We report it because a section about selective reporting cannot practise it, and we do not build on it.

Praise on tokens likewise loses significance at stage 1 (p = 0.17), which §4.7’s turn-count table does not show.

7.6 Claims this study withdrew about itself

Instability is one failure mode; analysis error is another. We discuss four here, chosen because each changed a substantive claim. They are not the complete list — the arithmetic and estimator errors are recorded in §2.4, §2.2 and `RESULTS.md`, and include a double-counted token total, an un-clustered confidence interval roughly 35% too narrow, a mislabelled “monotonic decline”, a false-positive claim that compared two different tests, a pooled inert cell that diluted every seven-level effect by about a third, and an unpaired-versus-paired estimator mix-up.

A timing effect that was selection. An early micro-experiment split the interruption’s reasoning-token effect by injection turn and found +21% at turn 1 against +39% at turn 2, $p = 0.004$. The injection turn was drawn at random, and a turn-2 injection can only fire in a trajectory that reaches turn 2 — the harder, longer tasks. The two *positions* were therefore measured on different task populations, 50 against 35. Recomputed on a control-defined population of 29 tasks: +20.0% against +29.1%, $p = 0.32$. Withdrawn. This is why injection turn is crossed rather than sampled (§2.2) and why the comparison population is fixed from control (§2.5).

A timing effect that was a proxy. The crossed run then found the interjection inert at turn 0 and large at turns 1 and 2, which looks like position. §4.6 shows it is not: holding turn index fixed and varying whether a candidate answer exists changes the effect from +0.34 to +1.97, while holding the moderator fixed and varying the turn index changes it from +1.97 to +1.66.

A ceiling explanation we tested and withdrew. When the continue signal’s progress effect looked like one hit between two nulls, the natural reading was that the 10-turn ceiling had truncated real progress. Truncating the ceiling-20 run’s own analysis to ten turns left the effect intact, so the ceiling explained almost none of the gap. The correction was written, then checked, then withdrawn — and §7.4 now shows the gap itself was the instrument.

A productive window that was a rate on a small denominator. A per-turn improvement rate of 1–5% at turn indices around 10–14 was read as evidence that the productive window extended that far. The rate was real; the conclusion was not. In absolute terms those turns contain almost no improvements — 98.3% of trajectories have reached their best answer by turn 10, at a median turn of 2 (§6.6).

7.7 Two infrastructure failures worth recording

Neither changed a conclusion; both are the kind of thing that corrupts a result without announcing itself.

Shared scratch directories. Two runs used the same tone over the same 50 tasks and the same 8 trials, and the scratch path did not include the run label — so 800 trajectories shared 400 execution directories, and two concurrent processes unlinked and rewrote one another’s output workbook. Every pass/fail flag in those runs was suspect. Token counts come from the provider’s API response and never touch the filesystem, so the cost effect was never in question. The grades were recovered by regrading from the archived logs, at no cost. Scratch paths are now namespaced by run tag and injection turn.

A silent range-parser bug. "P2:P7" parsed as sheet P2: plus cell P7, grading a one-cell range on a sheet that does not exist. Every affected lookup returned **None**, which reads as *unmeasurable* rather than as *broken*. It was caught by validating the regrade against a real task before scaling up, and it is why §6.2 describes that validation step.

7.8 What we would ask of this literature

Nothing here is a new methodological proposal; it is the discipline [22], [19] and [15] already ask for, applied to a study small enough that applying it is cheap. Three things, in order of what they cost.

1. **Report the estimator.** The unpaired difference of arm means and the paired task-clustered difference give different numbers for the same contrast — ours differed by up to 6% on the praise contrast (§2.4). A paper that does not say which it used cannot be compared with one that does. This is the defensible complaint about [5]: its paired t-tests treat ten runs as the unit and ignore clustering within the 50 items.

2. **Measure at least one contrast twice, on the same instrument.** It cost us nothing we would not have spent anyway, and it changed what we were willing to claim about four results: the continue-signal progress effect, the accuracy movement that accompanied it, the praise magnitude, and the timing effect of §7.6.
3. **Report ranges where you have them and say so where you do not.** We quote a range wherever a quantity was measured more than once and flag the single measurements as single (§8.6). [19] recommend this while also noting that single-format evaluation “may still be sufficient for many use cases”, so it is a reporting discipline rather than a design requirement.

The first is what the design this paper is positioned against lacked. The second is what its own authors eventually supplied, re-running the experiment the following spring and not reproducing its headline contrast on GPT-4o [6]. Their re-run is not a null — §1.1 and §3.3 give it in full — but the 4.0-point polite-to-rude gap it was known for becomes 0.4.

8. Limitations, and what we expect to generalise

8.1 The short version

Two models, one benchmark, fifty tasks, one agent scaffold. Everything in this paper is a claim about what these agents did on these tasks. We separate below what we expect to survive a change of setting from what we do not, because those are different bets and a reader should be able to disagree with them separately.

8.2 What we expect to generalise

The dissociation. That demand without evaluative language produces the effect, that negative register without demand does not, and that positive register without demand produces the opposite effect (§4.9) is a claim about what part of a message an agent acts on. Two of its three legs held on a second model from a different lab with very different stopping behaviour; the praise leg was directionally consistent but below power there (§4.7, §8.3). And in the two published stimulus sets whose per-condition outcomes can be inspected, the demand-bearing conditions are the ones that moved (§3.6). We would be surprised if it were specific to spreadsheets.

That register moves length, not correctness at the size the tone literature reports. This is already published for single-turn settings [10, 24], and our contribution is to replicate it where the moved quantity is *steps* rather than tokens within one response. Two independent lines of evidence converging is a reasonable basis for expecting it elsewhere. Note the bound, and that it is a bound on the *pooled* estimates: a single contrast here cannot exclude a four-point effect (§2.7), while the pooled contrasts of §2.8 can. Neither excludes an effect of a point or two.

Closing cues as a termination lever. This is the most distinctive result (§5) and the one whose generalisation we hold most loosely, because it rests on one run on one model. What the data support is an ordering — explicit close, conventional close, no signal, explicit continue — monotone in how strongly the message projects an end to the exchange. The mechanism behind that ordering is not demonstrated, and the one implication of a pre-closing account that we could test failed (§5.4). What travels, if anything does, is that an agent’s stopping behaviour is movable by a cue carrying no information about the task.

8.3 What we do not expect to generalise

Every magnitude. §7 is explicit: point estimates in this study span $1.5\times$ to $3.4\times$ across re-measurements of the same contrast on the same model, most of it across different turn ceilings. At matched ceiling the two larger effects reproduce within 10% while the smallest is marginally inconsistent with a common value (§7.3). Quoting any number here as *the* size of a register effect would repeat the error the paper is about.

The waste. That a continue signal buys mostly repeated work is, on our own data, a property of the model rather than of the signal. The Luna control produces 1.51 redundant steps to GLM’s 0.26, and a demand interjection adds +1.03 to Luna against +0.19 to GLM (§4.7). A model that stops early has little repetition available to be pushed into. Expect the *direction* to hold and the wastefulness to scale with how inclined the model already was to continue.

The praise magnitude specifically. GLM’s praise arm does not reach significance on its own ($p = 0.062$) and is below 80% power at its observed size (§2.7). The replication of §4.7 rests on the demand arm.

8.4 The substrate

SpreadsheetBench was chosen because it is execution-grounded — the grade comes from running the agent’s code against real test cases, not from a model’s judgement — and because its tasks are real workbook problems rather than synthesised ones. That grounding is the benefit; three costs follow from it.

Accuracy is structurally underpowered here. 26 of our 50 tasks are never solved by the control arm and 4 are always solved, so for more than half the sample the per-task accuracy difference is pinned at zero whatever the manipulation does (§2.7). This is why the accuracy null is stated as an equivalence against a pre-specified bound rather than as a measured zero, and why turn count is the primary outcome.

The turn ceiling is ours, not the benchmark’s. SpreadsheetBench’s official multi-round protocol caps at five rounds; we used ten and then twenty. §6.6 defends twenty empirically — the median turn at which the best answer is first reached is 2, and 98.3% of trajectories have peaked by turn 10 — but the choice is ours and comparisons to the benchmark’s published numbers should not be made across it.

The evaluator has a floor. Its own audit reports a 4% instruction-level false-negative rate and a 3.8% test-case-level false-omission rate [13]. Our per-turn regrade inherits it (§2.10).

A second substrate is the obvious next step and we have not run it. AppWorld [21] is the candidate: its stock evaluator keeps per-requirement pass and fail lists, so a per-turn “fraction of unit tests passed” measure can be computed without modifying it — the same instrument as §6, on a different task family, at roughly \$0.70 per trajectory. One piece of engineering it would require: what that benchmark treats as collateral damage is not a separate metric but additional entries in the same test battery, so separating it is work.

8.5 The scaffold

One agent loop, ReAct with code execution and execution feedback. Three properties of it are load bearing and a different scaffold could change any of them.

Each turn receives the original input, never the previous turn’s output. This is what makes turns independent attempts and makes “did attempt $k+1$ land closer?” well posed (§6.2).

An agent that accumulates state would need a different progress measure, and might well show real improvement across turns where ours does not.

The interjection is delivered as an execution observation. It arrives where tool output arrives, not as a fresh user turn. Whether an agent responds the same way to a message in the user channel is untested and is a plausible moderator of the closing-cue result in particular.

Turn 0 is inspection. On Luna, 98% of first turns run code and 0% produce a candidate answer; on GLM, 90% and 5% (§4.6, §4.7). The finding that an interjection is inert until work exists to be about is, on our data, a finding about this scaffold’s opening move.

8.6 Things we measured once

Because §7 treats a single measurement as a draw:

- The **bare closing cue, praise-the-work, and the Q4–Q5 praise isolation** (§5) — one run, one model. Two arms of that run were later re-measured and replicated in direction (§7.2); the three that carry the section’s argument have not been.
- The **opening-tone null** (§2.6) — one run, and on an instrument the paper itself declares confounded: its seven wrappers were not length-matched, and its reference level carried an extra task instruction that triples the rate of zero-turn trajectories (§2.2). The length-matching fix was applied only as a 150-trajectory re-run of that one arm; the seven-wrapper scale was never re-run matched. Its interval excludes a 44%-scale effect, which is stronger than a bare null, but the instrument is the weakest in the study.
- The **timing-is-a-proxy result** (§4.6) — one run, one arm, and its no-answer cells are small (31 tasks at turn 1, 8 at turn 2).
- **GLM**, on everything. One run at one ceiling.

8.7 Objections we expect — three we cannot answer, one we now can

One stimulus per construct. Every construct in this paper — demand, praise, insult, closing cue, “work remains” — is a single 28-token sentence. We cite [19] and [16] for prompt-format sensitivity and then generalise from one sentence per condition. A stimulus-sampling design, several paraphrases per construct, is the right version of this experiment and we did not run it. The length-matching and the shared **Checking in.** stem control *within* a comparison; they do nothing about whether our one praise sentence is representative of praise.

The demand coding is ours and post hoc. §4.2’s coding of the seven registers was made after seeing the effects — we say so there — and §3’s coding of other papers’ stimuli has no second rater. The probe of §4.3 is what tests the construct rather than asserting it, but it tests it on texts we wrote to embody our own coding.

This is the objection we would raise first against this paper, and it is not answerable by rewriting. The sharpest form of it is that the dissociation may be near-tautological: if “demand” means *contains an instruction* and “register” means *contains none*, then “instructions move behaviour and decorative text does not” is unsurprising and needs no manners-versus-demand framing. Three things would settle it, none of which we have done and all of which need new data rather than new prose:

1. **A blinded second coder** on both the seven-register set and §3’s cross-paper stimuli, with a reliability statistic reported. One rater is not a measurement.
2. **A demand scale fixed before outcomes are seen** — human or held-out-model ratings of “how strongly does this text demand continued work,” collected against stimuli whose turn counts

the raters cannot see.

3. **Several paraphrases per construct**, so the dissociation is shown to survive the wording and not just our wording. This is the same gap as the first objection above.

Until then the three-way dissociation in §4.9 should be read as *consistent with* a demand account rather than as establishing one over an instruction-following account.

The primary outcome is not corrected for multiplicity — now answered, and the guess in this paragraph was half wrong. An earlier draft of this objection said the load-bearing turn-count p-values “would not all” survive correction. §2.6 now corrects them, over the same 22-contrast family as the accuracy analysis, and **17 of 22 survive Benjamini–Hochberg at $q = 0.05$** . Of the four p-values this paragraph named, three survive — stage-1 Luna praise (0.019), L2 very polite (0.0055), L6 very rude (0.013) — and one does not: **GLM praise at 0.058**. The other four failures are the three insult contrasts and L5_rude, which is the dissociation §4 argues for. Correcting a primary outcome after the fact is still weaker than pre-registering the family, and we did not pre-register it.

Run-level drift is not excluded as an explanation of §7. Provider drift was checked *within* the opening run ($r = -0.003$ with elapsed time) but not *between* runs, which span three days. The roster is provider-pinned and the served provider was asserted on every call, but temperature 0 is not deterministic on any roster model (§2.2), and we cannot separate sampling variation from drift for the one matched-ceiling pair that disagrees (§7.3).

8.8 What we did not test

Whether any of this affects a human’s experience of the agent. We measured what the agent did. Whether users prefer, trust, or are better served by an agent that persists is a different question with a different method.

Whether the effects compound. Every run delivers exactly one interjection. An agent nagged repeatedly, or praised repeatedly, might habituate, escalate, or neither.

Whether register in the *system* prompt behaves like register mid-task. Our opening-tone null (§2.6) is about a register-varied instruction at the top of the user turn, not about a persona set in the system prompt, which is the configuration most deployments actually use.

The cost of being interrupted at all. Every crossed run compares an interjection against a *neutral interjection*, so the effects throughout are register effects, not interruption effects. Only the micro-experiment measured interruption itself — 3.47 to 4.11 turns under a neutral interruption, +51 reasoning tokens, $p = 0.15$ — once, on one model.

Two of the four roster models. DeepSeek and Qwen passed the $n=100$ validation gate and were never run. “Two models from different labs” describes what the budget reached, not a design choice; the four-model roster was specified and then not executed.

Whether any of this holds when the other party is adversarial. Every task here is solitary: the agent works on a workbook, and the only other voice in the transcript is ours. Negotiation is the obvious contrast, and it is now instrumented — TERMS-Bench sets out to diagnose negotiation agents beyond deal rate [25]. Whether a closing cue curtails a negotiating agent the way it curtails a working one, or whether a counterpart’s register does what a user’s does, is untested here and would need an outcome measure of that kind rather than ours.

Anything about refusals or safety behaviour. There were zero refusals in the opening-tone run’s 1,050 trajectories under every register including threatening, and the model never referred to the user’s tone across its 4,647 calls. Refusal counts for the interjection runs are recorded but were not analysed. That is an observation, not a test.

8.9 Summary

The result we are most confident in is a negative one: **rudeness without a demand does not make an agent work harder, and politeness without a demand does not either**. The arms that did make it work harder were the ones that told it to keep going, in whatever register. What moves an agent is being told to keep going, and what stops it is being told, in any of several ways, that the conversation is over. The strongest practical implication is a hazard rather than a technique — a pleasantry can end an agent’s work — and the strongest methodological implication is that this literature’s effect sizes should be treated as ranges until someone measures them twice.

References

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- [2] Ziv Ben-Zion, Zohar Elyoseph, Tobias Spiller, and Teddy Lazebnik. Inducing state anxiety in LLM agents reproduces human-like biases in consumer decision-making. *npj Artificial Intelligence*, 2(1):55, 2026. doi: 10.1038/s44387-026-00122-1.
- [3] Hanyu Cai, Binqi Shen, Lier Jin, Lan Hu, and Xiaojing Fan. Does tone change the answer? evaluating prompt politeness effects on modern LLMs: GPT, Gemini, and LLaMA, 2025. Verified by title and authors only; not read in full.
- [4] Alejandro Cuadron, Dacheng Li, Wenjie Ma, Xingyao Wang, Yichuan Wang, Siyuan Zhuang, Shu Liu, Luis Gaspar Schroeder, Tian Xia, Huanzhi Mao, Nicholas Thumiger, Aditya Desai, Ion Stoica, Ana Klimovic, Graham Neubig, and Joseph E. Gonzalez. The danger of overthinking: Examining the reasoning-action dilemma in agentic tasks, 2025. Unrefereed preprint.
- [5] Om Dobariya and Akhil Kumar. Mind your tone: Investigating how prompt politeness affects LLM accuracy (short paper), 2025. Unrefereed preprint.
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- [7] Vishal Gandhi and Sagar Gandhi. Prompt sentiment: The catalyst for LLM change, 2025. Unrefereed preprint.
- [8] Minyang Hu, Bo Yang, Zhinuo Zhou, Jiachen Liang, Jiahao Guo, Yiyang Yin, and Xiongwei Han. Redundant or necessary? a benchmark for detecting redundant steps in agent trajectories, 2026. Unrefereed preprint.
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- [14] Lennart Meincke, Ethan Mollick, Lilach Mollick, and Daniel Shapiro. Prompting science report 3: I’ll pay you or i’ll kill you — but will you care?, 2025. Unrefereed technical report.
- [15] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations, 2024. Unrefereed preprint (Anthropic).
- [16] Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. State of what art? a call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*, 12:933–949, 2024. doi: 10.1162/tacl_a_00681.
- [17] Thamilvendhan Mунirathinam. Will the agent recuse, and will it stop? measuring LLM-agent compliance with in-band governance signals at the access door and mid-flight, 2026. Verified by title and author only; not read in full.
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- [20] Moran Sun, Tianlin Li, Yuwei Zheng, Zhenhong Zhou, Aishan Liu, Xianglong Liu, and Yang Liu. How emotion shapes the behavior of LLMs and agents: A mechanistic study, 2026. Proposes the E-STEER emotion-steering framework. Unrefereed preprint.
- [21] Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. Appworld: A controllable world of apps and people for benchmarking interactive coding agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.850. Best Resource Paper Award.
- [22] Laurène Vaugrante, Mathias Niepert, and Thilo Hagendorff. A looming replication crisis in evaluating behavior in language models? evidence and solutions, 2024. Unrefereed preprint.
- [23] Sarel Weinberger and Amir Hozes. Prompt-induced waste in coding agents: Reasoning, effort, harness design, and end-to-end cost, 2026. Unrefereed preprint; cite a specific version — see comment.

- [24] Ziqi Yin, Hao Wang, Kaito Horio, Daisuke Kawahara, and Satoshi Sekine. Should we respect LLMs? a cross-lingual study on the influence of prompt politeness on LLM performance. In *Proceedings of the Second Workshop on Social Influence in Conversations (SICon 2024)*, pages 9–35. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.sicon-1.2.
- [25] Erica Zhang, Fangzhao Zhang, Aneesh Pappu, Batu El, Jose Blanchet, Susan Athey, Jiashuo Liu, and James Zou. TERMS-Bench: Diagnosing LLM negotiation agents beyond deal rate, 2026. Unrefereed preprint.
- [26] Tian Zhang and Meng Li. Should I be polite to my LLM relevance judge? tone as a severity operating-point shift, 2026. Verified by title and authors only; not read in full.
- [27] Jian Zhu, Yuzheng Zhang, Zeyao Ma, Bohan Zhang, Armin Schoepf, Daniel Woloch, and Peter Yiliu Wang. SpreadsheetBench 2: Evaluating agents on end-to-end business spreadsheet workflows, 2026. Verified by title and authors only; not read in full.

9. References

Most entries below are transcribed from `Literature review/01-sources.md`, which records what was verified and how. The **Verified** column reports that review’s own tag: *full text* means the paper was read in full and its quoted numbers checked against it; *abstract only* means it was not. Both of the entries that once needed a pass against the primary PDF have had it; §3.7 records what each check found.

Seven entries postdate that review and are not in it. Three — [2], [20] and [25] — were added after a citation audit. Four more — [3], [26], [17] and [27] — were surfaced by a simulated review panel and are listed in §3.8. Each of the seven is tagged at the read depth it actually has, and for the last four that is **title and author metadata only**. That is a much weaker standard than the rest of this list meets. It is marked rather than smoothed over, and **no claim anywhere in this paper rests on any of the seven**; the four in §3.8 are recorded as work that must be read before the novelty claim in §1.2 is submitted anywhere.

Venue is stated for each, because roughly half of this literature is unrefereed preprints and that bears on how much weight a reader should give it.

[benzion-2026] Ziv Ben-Zion, Zohar Elyoseph, Tobias Spiller, Teddy Lazebnik. “Inducing state anxiety in LLM agents reproduces human-like biases in consumer decision-making.” *npj Artificial Intelligence* 2(1), article 55, 23 Jun 2026. DOI 10.1038/s44387-026-00122-1. *Refereed journal*. — **Verified: abstract and publisher metadata only; not read in full.** Three models on a budget-constrained grocery task, 2,250 runs, before and after anxiety-inducing narratives; reports lower Basket Health Scores when primed ($\Delta = -0.081$ to -0.126 , $d = -1.07$ to -2.05). Cited in §1.2 as evidence that emotional context reaches an agent’s *actions*, which we treat as established rather than as our contribution. Distinct from our design on three counts stated there: the prime precedes the task, it is narrative rather than register, and the outcome is a preference score rather than a verifiable ground truth. [!] The effect sizes above are quoted from the abstract; anyone leaning on them should read the paper.

[balachandran-2025] Vidhisha Balachandran, Jingya Chen, Lingjiao Chen, Shivam Garg, Neel Joshi, Yash Lara, John Langford, Besmira Nushi, Vibhav Vineet, Yue Wu, Safoora

Yousefi. “Inference-Time Scaling for Complex Tasks: Where We Stand and What Lies Ahead.” arXiv:2504.00294, 31 Mar 2025. *Unrefereed preprint (Microsoft Research)*. — **Verified:** full text. Cited in §6.5 for oracle-assisted selection in inference-time scaling.

[cai-2025-tone] Hanyu Cai, Binqi Shen, Lier Jin, Lan Hu, Xiaojing Fan. “Does Tone Change the Answer? Evaluating Prompt Politeness Effects on Modern LLMs: GPT, Gemini, and LLaMA.” arXiv:2512.12812. *Unrefereed preprint*. — **Verified: title and authors against the arXiv abstract page; NOT read.** Surfaced by the review panel (§3.8). Directly on topic for §3 and absent from that section’s audit. Present in `Literature review/01-sources.md` but never cited.

[cuadron-2025] Alejandro Cuadron, Dacheng Li, Wenjie Ma, Xingyao Wang, Yichuan Wang, Siyuan Zhuang, Shu Liu, Luis Gaspar Schroeder, Tian Xia, Huanzhi Mao, Nicholas Thumiger, Aditya Desai, Ion Stoica, Ana Klimovic, Graham Neubig, Joseph E. Gonzalez. “The Danger of Overthinking: Examining the Reasoning-Action Dilemma in Agentic Tasks.” arXiv:2502.08235, 12 Feb 2025. *Unrefereed preprint*. — **Verified:** full text. Cited in §5.4 and §6.7 for *premature disengagement*, a term we deliberately decline to borrow.

[dobariya-kumar-2025] Om Dobariya, Akhil Kumar. “Mind Your Tone: Investigating How Prompt Politeness Affects LLM Accuracy (short paper).” arXiv:2510.04950, 6 Oct 2025. 5 pages. *Unrefereed preprint; comments field says submitted to Findings of ACL 2025*. Penn State. — **Verified:** full text. This is the paper the study is positioned against. **§3.7 item 2 is now closed:** the prefix pool for its 50-question dataset is printed in [6] Table 1 and gives two or three variants per level, not one. The check went against our reading and §3.3 is weakened accordingly. Residual gap recorded in §3.3: we have seen the 2026 paper’s table for that dataset, not the 2025 paper’s own.

[dobariya-kumar-2026] Om Dobariya, Akhil Kumar. “Mind Your Tone: Does Tone Alter LLM Performance?” arXiv:2605.29027, 27 May 2026. *AMCIS 2026 (Thirty-second Americas Conference on Information Systems, Reno)* — *refereed*. — **Verified: full text, primary PDF, read end to end.** The same authors’ own re-run: four models, two datasets (the original 50 questions at five tones; 570 MMLU questions at seven), ten runs each. Source of §3.3’s complete prefix pool and of the shared-preamble brevity instruction. Reports no token counts — do not confuse it with [10], which is a different paper. Cited in §1.1, §3.3 and §7.8.

[gandhi-2025] Vishal Gandhi, Sagar Gandhi. “Prompt Sentiment: The Catalyst for LLM Change.” arXiv:2503.13510, 14 Mar 2025. *Unrefereed preprint*. — **Verified:** full text. Reports that positive prompts *increase* verbosity — essay and blog responses 8.1% longer than neutral — on single-turn tasks across five models. Cited in §5.4, where we state plainly that their dependent variable (verbosity of one response) is not ours (steps taken on a task), so their result and our praise effect are not in contradiction.

[huang-2024] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, Denny Zhou. “Large Language Models Cannot Self-Correct Reasoning Yet.” arXiv:2310.01798, 3 Oct 2023 (rev. 14 Mar 2024). *ICLR 2024* — *refereed*. — **Verified:** full text. Cited in §6.5.

[kumar-dobariya-2026] Akhil Kumar, Om Dobariya. “Understanding Tone-Dependent Inference Cost in Large Language Models.” arXiv:2607.23915, 27 Jul 2026. 25 pages. *Unrefereed preprint*. — **Verified: full text, primary PDF, read end to end.** Source of the *social register* definition this paper adopts, and of §3.2’s two tables — their Table 2 (all seven prefixes, word counts, VADER scores) and Table 3 (accuracy and output tokens, four models, ten runs). **§3.7 item 1 is closed against this PDF, and it corrected us:** their Neutral prefix asks for “the single letter”, making it the most explicit output-length instruction of the seven rather than the no-instruction baseline our draft called it. Distinct from [6]: different arXiv number, author order, and dependent variable (inference cost, not accuracy).

[lakens-2017] Daniel Lakens. “Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses.” *Social Psychological and Personality Science* 8(4), 2017, 355–362. DOI 10.1177/1948550617697177. *Refereed journal*. — **Verified:** full text. The source for §2.8’s TOST framing. Cited for equivalence testing and **not** [15], which contains none.

[li-2023] Cheng Li, Jindong Wang, Yixuan Zhang, Kaijie Zhu, Wenxin Hou, Jianxun Lian, Fang Luo, Qiang Yang, Xing Xie. “Large Language Models Understand and Can be Enhanced by Emotional Stimuli.” arXiv:2307.11760, 14 Jul 2023 (v7, 12 Nov 2023). Short version at LLM@IJCAI’23. *Unrefereed preprint*. — **Verified:** full text. EmotionPrompt; source of §3.4’s stimulus table.

[ma-2024] Zeyao Ma, Bohan Zhang, Jing Zhang, Jifan Yu, Xiaokang Zhang, Xiaohan Zhang, Sijia Luo, Xi Wang, Jie Tang. “SpreadsheetBench: Towards Challenging Real World Spreadsheet Manipulation.” arXiv:2406.14991, 21 Jun 2024 (rev. Oct 2024). *NeurIPS 2024 Spotlight — refereed*. — **Verified:** full text. Our substrate. Source of the five-round official protocol (§2.1) and the evaluator audit (§2.10).

[meincke-2025-report3] Lennart Meincke, Ethan Mollick, Lilach Mollick, Dan Shapiro. “Prompting Science Report 3: I’ll pay you or I’ll kill you — but will you care?” arXiv:2508.00614, 1 Aug 2025. *Unrefereed technical report*. — **Verified:** full text. Cited in §3.5. Note their comparisons are uncorrected for multiplicity, which §3.5 states.

[miller-2024] Evan Miller. “Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations.” arXiv:2411.00640, 1 Nov 2024. Anthropic. *Unrefereed preprint, widely adopted*. — **Verified:** full text. Source of the clustered-standard-error treatment and the MDE inversion (§2.4, §2.7). **Contains no equivalence framework** — see [11].

[mizrahi-2024] Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, Gabriel Stanovsky. “State of What Art? A Call for Multi-Prompt LLM Evaluation.” *Transactions of the Association for Computational Linguistics* 12 (2024), 933–949. DOI 10.1162/tacl_a_00681. *Refereed journal*. — **Verified:** full text. 5,000 manually verified instruction paraphrases (more than 175 per task) evaluated over 6.5M instances, 20 LLMs, 39 tasks, 3 benchmarks. Kendall’s W across prompts is below 0.85 for most tasks, so **model ranking**, not just absolute score, depends on the template. [!] **Scope their recommendation carefully:** it is use-case-dependent, not a blanket call for ranges. They recommend averaging across many prompts when measuring robustness, but comparing models on “their corresponding top-performing prompt” when developing a downstream system, and they propose a Combined Performance Score uniting the two. §2.3 and §8.7 cite them for the first use case, which is ours. They also find automatically generated paraphrases sufficient for their metrics “without having to manually verify them” — relevant to §8.7’s stimulus-sampling objection, which we do not answer.

[munirathinam-2026-recuse] Thamilmvendhan Munirathinam. “Will the Agent Recuse, and Will It Stop? Measuring LLM-Agent Compliance with In-Band Governance Signals at the Access Door and Mid-Flight.” arXiv:2606.06460. *Unrefereed preprint*. — **Verified: title and author against the arXiv abstract page; NOT read.** Surfaced by the review panel (§3.8). [!] **On its title alone this is the nearest neighbour to §5’s result** — mid-flight signals to an agent, and whether it stops. §5.1 claims no prior work connects closing-sequence structure to agent termination. That claim cannot stand until this paper is read.

[redundancybench] Minyang Hu, Bo Yang, Zhinuo Zhou, Jiachen Liang, Jiahao Guo, Yiyang Yin, Xiongwei Han. “Redundant or Necessary? A Benchmark for Detecting Redundant Steps in Agent Trajectories.” arXiv:2605.29893, 28 May 2026. *Unrefereed preprint; the anonymised code repository indicates it is under double-blind review*. — **Verified:** full text. Source of the term *redundant step* (§2.3). We do not compare base rates against theirs, as §2.3 states.

[schegloff-sacks-1973] Emanuel A. Schegloff, Harvey Sacks. “Opening up Closings.” *Semiotica* 8(4): 289–327, 1973. DOI 10.1515/semi.1973.8.4.289. *Refereed journal*. — **Verified:** canonical text. The source for pre-closing structure (§5.1).

[sclar-2024] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, Alane Suhr. “Quantifying Language Models’ Sensitivity to Spurious Features in Prompt Design, or: How I learned to start worrying about prompt formatting.” arXiv:2310.11324, 17 Oct 2023 (rev. Jul 2024). *ICLR 2024* — *refereed*. — **Verified:** full text. Cited for the **median 7.5-point** format spread used as §2.7’s comparator. Their widely-quoted 76 points is a single-task maximum on LLaMA-2-13B and is explicitly a lower bound; we do not use it.

[sun-2026-esteer] Moran Sun, Tianlin Li, Yuwei Zheng, Zhenhong Zhou, Aishan Liu, Xianglong Liu, Yang Liu. “How Emotion Shapes the Behavior of LLMs and Agents: A Mechanistic Study.” arXiv:2604.00005, 9 Mar 2026, v1. 15 pages. *Unrefereed preprint*. [!] The date looks wrong for the identifier — a 2604 prefix normally denotes an April 2026 submission — but 9 March is what both primary sources state: the arXiv abstract page reads “[Submitted on 9 Mar 2026]” with “[v1] Mon, 9 Mar 2026”, and OpenAlex independently gives 2026-03-09. We record what the sources say rather than the date the numbering convention implies. — **Verified: abstract and registry metadata only; not read in full.** Proposes **E-STEER**, an emotion-steering framework that embeds emotion as a controllable variable in hidden states; reports non-monotonic emotion-behaviour relations and that specific emotions “systematically shape multi-step agent behaviors.” Cited in §1.2 for that last point. [!] Note the channel: this is a **representation-level** intervention, not a prompt manipulation, so it bounds our claim without competing with it. [!] It does **not**, on the abstract, compare the magnitude of agent-level variation against single-step variation; do not cite it for that without checking the full text.

[trivedi-2024] Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, Niranjan Balasubramanian. “AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents.” arXiv:2407.18901, 26 Jul 2024. *ACL 2024 (2024.acl-long.850)* — *refereed, Best Resource Paper*. — **Verified:** full text plus evaluator source. Cited in §8.4 as the second substrate we did not run.

[vaugrante-2024] Laurène Vaugrante, Mathias Niepert, Thilo Hagendorff. “A Looming Replication Crisis in Evaluating Behavior in Language Models? Evidence and Solutions.” arXiv:2409.20303, 30 Sep 2024. *Unrefereed preprint*. — **Verified:** full text. The template for §7. Note their own design is single-run at temperature 0, which §7.1 states.

[weinberger-hozez-2026] Sarel Weinberger, Amir Hozez (PointFive). “Prompt-Induced Waste in Coding Agents: Reasoning, Effort, Harness Design, and End-to-End Cost.” arXiv:2608.01347. *Unrefereed preprint, genuinely preregistered*. — **Verified:** full text, v1 and v6; title and six-version history confirmed against the arXiv abstract page, which dates v6 to 10 Sep 2026. The subtitle is v6’s; v1 carried the short title alone. **Cited at v6 (10 Sep 2026) throughout.** [!] The paper has six versions and grew roughly 8× between v1 and v6; v6 reports 4,644 valid runs and removes several figures and quotes present in v1, so a version must be named when citing it. Cited in §4.5 for the tokens-per-step versus steps distinction, and in §5.5, where their **bounded_efficiency** arm establishes that a stop condition in the *opening* prompt reduces agentic work — a result we do not claim.

[zhang-2026-politejudge] Tian Zhang, Meng Li. “Should I Be Polite to My LLM Relevance Judge? Tone as a Severity Operating-Point Shift.” arXiv:2609.09703. *Unrefereed preprint*. — **Verified: title and authors against the arXiv abstract page; NOT read.** Surfaced by the review panel (§3.8), which reported that it uses several independently written paraphrases per tone level across eight judge models — the stimulus-sampling design §8.7 concedes we did not run. We

have not confirmed that description against the paper.

[**zhang-2026-termsbench**] Erica Zhang, Fangzhao Zhang, Aneesh Pappu, Batu El, Jose Blanchet, Susan Athey, Jiashuo Liu, James Zou. “TERMS-Bench: Diagnosing LLM Negotiation Agents Beyond Deal Rate.” arXiv:2605.13909, 13 May 2026 (v2, 13 Jun 2026). *Unrefereed preprint*. — **Verified: title, authors and version history against the arXiv abstract page; not read in full.** Cited once, in §8.8, as the instrument a negotiation follow-up would need. We make no claim about its contents beyond what its title states, and we have not run it.

[**zhu-2026-spreadsheetbench2**] Jian Zhu, Yuzheng Zhang, Zeyao Ma, Bohan Zhang, Armin Schoepf, Daniel Woloch, Peter Yiliu Wang. “SpreadsheetBench 2: Evaluating Agents on End-to-End Business Spreadsheet Workflows.” arXiv:2606.29955. *Unrefereed preprint*. — **Verified: title and authors against the arXiv abstract page; NOT read.** Surfaced by the review panel (§3.8). A successor to our own substrate [13], sharing authors with it. §8.4 calls a second substrate “the obvious next step” without naming this one.

[**yin-2024**] Ziqi Yin, Hao Wang, Kaito Horio, Daisuke Kawahara, Satoshi Sekine. “Should We Respect LLMs? A Cross-Lingual Study on the Influence of Prompt Politeness on LLM Performance.” *Proceedings of the Second Workshop on Social Influence in Conversations (SICon 2024)*, ACL, pp. 9–35. DOI 10.18653/v1/2024.sicon-1.2. *Refereed workshop paper*. — **Verified: full text.** The largest- n study in this literature and the origin of the politeness-and-performance question.

Outstanding before submission

Seven items are open, all of the same kind. [2], [20] and [25] have been verified as existing — title, authors, venue and version history against the publisher record or the arXiv abstract page — but none has been read in full. Each is cited for a claim taken from its own abstract, and §1.2 and §8.8 are written so that nothing depends on a figure from any of them. Four further entries (§3.8) are weaker still — verified to exist by title and author, nothing more. Reading all seven is the remaining pre-submission task on this list, and the §3.8 four are the higher priority of the two groups, because one of them may occupy ground §5.1 claims is unoccupied. In particular, [20] must not be cited for a magnitude comparison between agent-level and single-step variation until its full text is checked; its abstract does not make one.

Both §3.7 primary-PDF checks are closed, against arXiv:2605.29027 and arXiv:2607.23915. Neither confirmed the draft as written, and both corrections are stated in place in §3.2 and §3.3 rather than quietly absorbed: the first strengthened the argument after exposing a mis-coding of ours, the second weakened a case we now rest nothing on.

Five earlier items are now closed. **§3.7 item 2** (arXiv:2605.29027): its Table 1 prints two or three variants per level for the 50-question dataset, not one — only two of six hostile variants carry a demand, and the previous draft quoted exactly those two, so §3.3 is rewritten and the case downgraded. **§3.7 item 1** (arXiv:2607.23915): their Table 2 is now quoted in full in §3.2; no polite prefix constrains output length, but the *Neutral* one does, which removes the anomaly our draft had conceded and makes the separation complete on all four models. [**chen-2023**] **has been dropped**: the words §6.5 quoted for it were Huang et al.’s prose about Self-Debug rather than a quotation from Chen et al., and the point is made through [9] alone. **[16] has been read in full**, and the entry above now scopes their recommendation correctly — it is use-case-dependent rather than a blanket call for ranges. **The two Weinberger & Hozes keys are merged**, cited at v6.